

Barnes Resolvents and Higher Harmonic Identities

Polynomial Hurwitz transforms, complex kernel powers,
quartic Stieltjes moments, and transverse Tornheim data

Research continuation prepared for Vladimir Reshetnikov

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Abstract

We continue the exact-identity programme of the ProveIt polylogarithm manuscript from a fixed repository snapshot. A polynomially weighted Hurwitz transform gives the individual cotangent resolvents of every unit-period multiple Gamma function with an explicit normalization. For the Barnes G function, the answer uses ordinary polylogarithms, their order derivatives, and one convergent depth-two derivative; an antisymmetric component reduces entirely to depth one. The same transform gives all weighted Stieltjes coefficients and their full endpoint corrections. We also develop complex powers of the kernel, an exact quartic digamma–harmonic bridge, and an independently convergent transverse coordinate for higher cyclic Tornheim derivatives. Every reduction is stated relative to an explicit function class; no arithmetic independence or literature-wide priority claim is made. Proofs, normalization checks, reproducible computations, a narrowly scoped correction to a published Barnes identity, and further research questions accompany the results.

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1 Scope, source baseline, and conventions

The starting point is the ProveIt repository at commit `7bd45777a8a127e5dc69aff8887ca36a79a3059d`, whose recorded time is 11 October 2026, 01:57:13 UTC. We obtained the canonical manuscript’s 81 chapter files, its README, validation record, preamble and references, and the text members of all eleven archives then present in `docs/incoming`. The source review was targeted at the identities used below; it was not a line-by-line soundness audit of every volume. The accompanying source inventory identifies the snapshot and the incoming Git blob identifiers.

The newest source, *Harmonic Parity and Resolvent Identities* [2], already settles the rational cotangent-resolvent problem, the balanced negapolygamma ladder, and a quartic directional normal form. Those conclusions are baseline results here. Its questions R4 and R6 instead ask for nonintegral complex kernel powers and individual multiple Gamma transforms. The cubic harmonic report [3] asks for a further bridge from higher digamma powers to harmonic Laurent coefficients. These are the principal targets of the present continuation.

Source target	Exact scope pursued in this article
Parity R6: individual Barnes functions	A finite formula for every polynomially weighted, unit-period multiple Gamma resolvent, including all normalization polynomials; an explicit G formula.
Parity R4: complex powers	A joint spectral transform and conversion at positive integer exponent crossings, with the branch fixed along the integration interval.
Cubic Q6: higher digamma powers	The fourth-power finite part in ordinary zeta/Stieltjes data and explicitly defined depth-two and depth-three harmonic Laurent coefficients; an absolutely convergent series representation.
Parity R10: a transverse cyclic coordinate	A convergent integral representative and its use in the fifth cyclic directional layer.

Table 1: Developments relative to the inspected reports. Theorems below state the precise domains; a reduction to higher-depth data is not a reduction to an unspecified algebra of familiar constants.

1.1 What remains a conjecture

The short Gaussian S_6 evaluation with label `cycloquot:conj:S6` and the revised S_8 evaluation with label `s8new:conj:S8` remain conjectural. The older frozen S_8 vector was rigorously rejected in the existing manuscript; it is a different candidate. None of the transformations in this article is an exact certificate for either current candidate. Similarly, the familiar cubic constants are not declared to belong to a smaller classical constant algebra simply because an integral or a new convergent series represents them.

1.2 Notation

We use the standard Hurwitz Laurent convention

$$\zeta(1-s, x) = -\frac{1}{s} + \sum_{m=0}^{\infty} \gamma_m(x) \frac{s^m}{m!}, \quad \gamma_0(x) = -\psi(x), \quad \psi = \frac{\Gamma'}{\Gamma}. \quad (1.1)$$

Thus $\gamma_m = \gamma_m(1)$ and $\gamma = \gamma_0$. An ordinary prime on ζ means a derivative in its first, spectral argument. A superscript on ψ or $\gamma_m(x)$ means an argument derivative. We write

$$\text{Li}_s^{[j]}(q) = \partial_s^j \text{Li}_s(q), \quad H_n^{(r)} = \sum_{k=1}^n k^{-r}, \quad H_n = H_n^{(1)}.$$

All logarithms of positive real numbers are real. Complex logarithms are fixed explicitly whenever they occur in a power. Products of positive real Gamma and Barnes functions on $(0, 1)$ use their real logarithms.

The endpoint finite part in this article is

$$\text{FP}_x \int_0^1 f(x) dx = [\delta^0 (\log \delta)^0] \int_\delta^{1-\delta} f(x) dx. \quad (1.2)$$

This definition applies when the integral has a finite asymptotic expansion of nonvanishing powers and logarithms plus a constant and a vanishing remainder. For complex exponents, the removed powers may oscillate as well as diverge. Our proofs either establish that expansion or give an equivalent explicit subtraction. A parameter Laurent constant, denoted CT, is a different operation. Their equality is asserted only after the relevant correction has been proved.

1.3 Classical inputs and the meaning of a contribution

The Hurwitz Fourier formula, the argument-shift law, Gamma derivatives, Bernoulli polynomial evaluations, and ordinary polylogarithm calculus are standard [4, 5]. Espinosa–Moll already developed Hurwitz integrals and balanced negapolygamma functions [7, 8]. The multiple-Gamma/Hurwitz-zeta conversion is classical [9, 10]. The Fourier coefficients of $x \log \Gamma(x)$ and $\log G(x)$ are also known, in particular in Mezö’s paper [11]. We do not reclassify any of those facts as discoveries.

Our Barnes result is a finite resolvent reduction, valid with an arbitrary polynomial weight and every specified normalization constant. Its spectral form supplies an all-index Stieltjes extension at the same time. The later sections isolate the precise additional harmonic and transverse coordinates rather than hiding them inside unnamed finite parts. These are developments relative to the pinned source programme. Numerical checks in the package support independent error detection; the proofs in the text carry the mathematical claims.

2 A polynomially weighted Hurwitz resolvent

2.1 The two Fourier directions and a convergent depth-two function

For $z \in \mathbb{C} \setminus \mathbb{R}$, put

$$\begin{aligned} \epsilon &= \text{sgn}(\Im z), \quad \omega = 2\epsilon\pi i, \quad \sigma = -\omega, \quad a = z + \epsilon\pi i, \\ q &= \frac{z - \epsilon\pi i}{z + \epsilon\pi i}, \quad c = \frac{2\epsilon\pi i}{a^2}, \quad |q| < 1, \\ \text{Log } \omega &= \log(2\pi) + \epsilon\pi i/2, \quad \text{Log } \sigma = \log(2\pi) - \epsilon\pi i/2. \end{aligned} \quad (2.1)$$

Let $K(x) = \pi \cot(\pi x)$ and $R_z(x) = (K(x) - z)^{-1}$. The elementary identity

$$R_z(x) = -\frac{1}{a} + c \sum_{n \geq 1} q^{n-1} e^{\omega n x} \quad (2.2)$$

converges uniformly with every argument derivative. In particular $R_z(0) = R_z(1) = 0$ and $R_z(x) = x + O(x^2)$ at zero.

The natural depth-two function for multiplication by a polynomial is

$$\mathcal{O}_{\alpha,\beta}(q) = \sum_{n,m \geq 1} \frac{q^n}{(m+n)^\alpha m^\beta} = \text{Li}_{\alpha,\beta}(q, q^{-1}). \quad (2.3)$$

Here our convention is $\text{Li}_{\alpha,\beta}(u, v) = \sum_{k > m \geq 1} u^k v^m / (k^\alpha m^\beta)$. The last expression in (2.3) is interpreted by the displayed convergent series, not by assigning separate independent branches to the two arguments.

Lemma 2.1 (Normal convergence). *The series (2.3), with all finite order derivatives, converges normally on*

$$|q| < 1, \quad \Re(\alpha + \beta) > 1.$$

It vanishes at $q = 0$, and division by q has a removable value there.

Proof. On a compact parameter set split the inner sum at $m = 2n$. For $m > 2n$, the summand is bounded by a fixed multiple of $|q|^n m^{-\Re(\alpha+\beta)}$. For $m \leq 2n$, all possible growth is bounded by a fixed power of n , and the geometric factor dominates it. Derivatives introduce only powers of $\log m$, $\log(m+n)$ and n , which are handled by a small margin in the strict inequalities. The same bounds justify termwise differentiation. Every term contains q^n with $n \geq 1$. $\square$

We shall also need the normally convergent subtracted functions

$$U_j(s, q) = \sum_{n,m \geq 1} q^n m^{-s} ((m+n)^{-j} - m^{-j}), \quad (2.4)$$

$$V_j(s, q) = \sum_{n,m \geq 1} q^n m^{-j} ((m+n)^{-s} - m^{-s}), \quad (2.5)$$

where $j \geq 1$. Each is holomorphic for $\Re s > -j$, $|q| < 1$. In the common initial domain they satisfy

$$U_j = \mathcal{O}_{j,s} - \frac{q}{1-q} \zeta(s+j), \quad V_j = \mathcal{O}_{s,j} - \frac{q}{1-q} \zeta(s+j). \quad (2.6)$$

For example, in the tail $m > 2n$ each difference gains a factor $O(n/m)$, uniformly on compact sets. The remaining part is geometrically summable as in Lemma 2.1. This proves the claimed larger domain directly, without assigning values to divergent unsubtracted summands.

2.2 The finite polynomial data

For a polynomial P of degree d , define

$$\mu_P = \int_0^1 P(x) dx, \quad b_j(P) = \frac{(-1)^{j-1}}{\omega^j} (P^{(j-1)}(1) - P^{(j-1)}(0)), \quad 1 \leq j \leq d. \quad (2.7)$$

The sum is empty when P is constant. Repeated integration by parts gives

$$\int_0^1 P(x) e^{\omega k x} dx = \begin{cases} \mu_P, & k = 0, \\ \sum_{j=1}^d \frac{b_j(P)}{k^j}, & k \in \mathbb{Z} \setminus \{0\}. \end{cases} \quad (2.8)$$

The derivative of order d is constant and hence contributes no endpoint difference. Thus only these finitely many data occur.

Define the elementary polynomial resolvent

$$T_P(z) = \int_0^1 P(x) R_z(x) dx = -\frac{\mu_P}{a} + \frac{c}{q} \sum_{j=1}^d b_j(P) \text{Li}_j(q). \quad (2.9)$$

All quotients by q in this article are given their removable values at $q = 0$, equivalently $z = \epsilon\pi i$.

2.3 The master identity

Set

$$\begin{aligned} \mathcal{B}_P(s, q) &= \sigma^{-s} \mu_P \text{Li}_s(q) \\ &+ \sum_{j=1}^d b_j(P) \left\{ \omega^{-s} U_j(s, q) + \sigma^{-s} (\text{Li}_s(q) \text{Li}_j(q) + (-1)^j V_j(s, q)) \right\}. \end{aligned} \quad (2.10)$$

Theorem 2.2 (Polynomial Hurwitz resolvent). *For every polynomial P , every $z \notin \mathbb{R}$, and $\Re s > 0$,*

$$I_P(s, z) := \int_0^1 P(x) \zeta(1-s, x) R_z(x) dx = \frac{c\Gamma(s)}{q} \mathcal{B}_P(s, q). \quad (2.11)$$

The bracket $\mathcal{B}_P$ is holomorphic on $\Re s > -1$. The formula therefore gives a meromorphic continuation through $s = 0$, with residue $-T_P(z)$ and no other pole in that half-plane. On $\Re s > 0$ it is a finite depth-two polylogarithmic formula by (2.6).

Proof. Initially take $\Re s > 1$. The classical Hurwitz Fourier expansion [4, §25.11] is absolutely convergent and reads

$$\zeta(1-s, x) = \Gamma(s) \sum_{m \geq 1} m^{-s} (\omega^{-s} e^{\omega m x} + \sigma^{-s} e^{-\omega m x}). \quad (2.12)$$

Multiply this by (2.2) and use (2.8). The constant Fourier term of R_z gives

$$-\frac{\Gamma(s)}{a} \sum_j b_j(P) (\omega^{-s} + (-1)^j \sigma^{-s}) \zeta(s+j).$$

The equal-frequency diagonal contributes $c\Gamma(s)\sigma^{-s}\mu_P \text{Li}_s(q)/q$. The same-sign frequencies contribute $\mathcal{O}_{j,s}$. For the remaining frequencies split at $m = n$:

$$\sum_{\substack{n, m \geq 1 \\ m \neq n}} \frac{q^n}{m^s (n-m)^j} = \text{Li}_s(q) \text{Li}_j(q) + (-1)^j \mathcal{O}_{s,j}(q). \quad (2.13)$$

Indeed $m < n$ is parameterized by $n = m + h$ and gives the product; $m > n$ is parameterized by $m = n + h$ and gives the second sum. Absolute convergence justifies these changes of variables.

Insert (2.6). Every ordinary zeta term cancels because

$$\frac{c}{1-q} = \frac{1}{a}. \quad (2.14)$$

What remains is exactly (2.11). Both sides are holomorphic on $\Re s > 0$, so the initial identity extends there. On the left, use $\zeta(1-s, x) = x^{s-1} + \zeta(1-s, x+1)$: the first term times PR_z is locally integrable for $\Re s > -1$, and the second has just the constant Hurwitz pole $-1/s$. The normal convergence of U_j, V_j proves the same assertion on the right. Finally, telescoping in m gives

$$U_j(0, q) = -\frac{\text{Li}_j(q)}{1-q}, \quad V_j(0, q) = 0.$$

Since $\text{Li}_0(q) = q/(1-q)$,

$$\mathcal{B}_P(0, q) = \mu_P \frac{q}{1-q} - \sum_j b_j(P) \text{Li}_j(q) = -\frac{q}{c} T_P(z). \quad (2.15)$$

This proves the residue, including the removable $q = 0$ case. $\square$

Example 2.3 (The first nonconstant weight). For $P(x) = x$, one has $\mu_P = 1/2$, $b_1 = 1/\omega$ and

$$\mathcal{B}_x(s, q) = \frac{1}{2} \sigma^{-s} \text{Li}_s(q) + \frac{1}{\omega} \{ \omega^{-s} U_1(s, q) + \sigma^{-s} (\text{Li}_s(q) \text{Li}_1(q) - V_1(s, q)) \}. \quad (2.16)$$

This is the missing $x \log \Gamma(x)$ input to an individual Barnes- G transform. Setting $P = 1$ deletes U_j, V_j and recovers the previously established unweighted formula.

2.4 All weighted Stieltjes coefficients

Theorem 2.4 (A holomorphic Stieltjes generator). *For every polynomial P , integer $m \geq 0$, and $z \notin \mathbb{R}$,*

$$\boxed{\int_0^1 P(x) \gamma_m(x) R_z(x) dx = \frac{m!c}{q} [s^{m+1}] \Gamma(1+s) \mathcal{B}_P(s, q).} \quad (2.17)$$

These are ordinary absolutely convergent integrals. Equivalently their exponential generating function is

$$I_P(s, z) + \frac{T_P(z)}{s} = \frac{c}{q} \frac{\Gamma(1+s) \mathcal{B}_P(s, q) - \mathcal{B}_P(0, q)}{s}. \quad (2.18)$$

Proof. The shift law $\gamma_m(x) = x^{-1} \log^m x + \gamma_m(x+1)$ and $R_z = O(x)$ prove convergence. The same bounds, locally uniformly for s near zero, allow (1.1) to be integrated after removing its constant pole. Now use Theorem 2.2 and (2.15). The numerator on the right of (2.18) vanishes at zero. $\square$

For instance, the first two moments are obtained from

$$\int_0^1 P \gamma_0 R_z dx = \frac{c}{q} (\partial_s \mathcal{B}_P(0, q) - \gamma \mathcal{B}_P(0, q)), \quad (2.19)$$

$$\int_0^1 P \gamma_1 R_z dx = \frac{c}{2q} (\partial_s^2 \mathcal{B}_P(0, q) - 2\gamma \partial_s \mathcal{B}_P(0, q) + (\gamma^2 + \zeta(2)) \mathcal{B}_P(0, q)). \quad (2.20)$$

Every derivative of the subtracted double sums is normally convergent. In particular these formulas do not require subtracting two divergent numerical evaluations of (2.6).

2.5 Argument derivatives and complete local corrections

Let

$$A_p(s) = \prod_{j=1}^p (s - j), \quad r_{P,p}(z) = [x^p](P(x)R_z(x)). \quad (2.21)$$

The empty product is one. Meromorphic continuation of I_P to any desired finite spectral neighborhood can be constructed by subtracting finitely many terms from the Taylor expansion of $P(x)R_z(x)$ at zero. This supplies an unambiguous meaning to the next formula even when $s - p$ is outside the normal domain of the first subtraction in (2.4)–(2.5).

Theorem 2.5 (Weighted argument-derivative conversion). *For $m \geq 0$, $p \geq 1$ and a polynomial P ,*

$$\boxed{\text{FP}_x \int_0^1 P(x) \gamma_m^{(p)}(x) R_z(x) dx = m! [s^m] \left\{ A_p(s) I_P(s - p, z) - \frac{r_{P,p}(z) A_p(s)}{s} \right\}}. \quad (2.22)$$

The expression in braces is holomorphic near $s = 0$. The entire polynomial amplitude $A_p(s)$ must be subtracted, not only its value at zero.

Proof. The Hurwitz argument ladder gives the meromorphic identity

$$\partial_x^p \zeta(1 - s, x) = A_p(s) \zeta(1 - s + p, x).$$

The singular part at zero is $A_p(s)x^{s-p-1}$. After multiplication by PR_z , the only exponent crossing -1 at $s = 0$ is $r_{P,p}A_p(s)x^{s-1}$. Its continued integral is $r_{P,p}A_p(s)/s$, whereas the finite part of every coefficient $x^{-1} \log^j x$ on $(0, 1)$ is zero. All other terms commute with coefficient extraction. To justify the latter statement, subtract a Taylor polynomial through degree $p + 1$ on $(0, \delta_0)$, integrate its terms explicitly, and use normal integrability of the remainder. The nonsingular Hurwitz tail at $x + 1$ is analytic after an argument derivative. This proves the formula and the holomorphy assertion. $\square$

For a completely explicit continuation of the depth-two expression, one can subtract N terms of $(1 + n/m)^{-j}$ and $(1 + n/m)^{-s}$. The resulting known terms are respectively

$$\sum_{k=0}^{N-1} \frac{(-1)^k (j)_k}{k!} \text{Li}_{-k}(q) \zeta(s + j + k), \quad (2.23)$$

$$\sum_{k=0}^{N-1} \frac{(-1)^k (s)_k}{k!} \text{Li}_{-k}(q) \zeta(s + j + k). \quad (2.24)$$

The remainders are normally convergent for $\Re s > 1 - j - N$. This follows by the same $m > 2n$ split and the finite Taylor remainder estimate; the terms $m \leq 2n$ are still geometrically summable. Thus (2.22) is a constructive finite calculation plus convergent sums at every fixed m, p .

Repeated resolvents follow by z differentiation:

$$\text{FP}_x \int_0^1 \frac{P(x) \gamma_m^{(p)}(x)}{(K(x) - z)^r} dx = \frac{1}{(r-1)!} \partial_z^{r-1} \text{FP}_x \int_0^1 P(x) \gamma_m^{(p)}(x) R_z(x) dx. \quad (2.25)$$

Local subtractions are holomorphic in z on compact subsets of either half-plane. If P vanishes to order e at zero, the integral in (2.25) is absolutely convergent when $e + r > p$.

3 Individual Barnes functions and all multiple Gamma ranks

The balanced functions in the preceding reports are particular linear combinations of Hurwitz-zeta derivatives and Bernoulli polynomials. An individual multiple Gamma function also carries polynomial coefficients in x . Theorem 2.2 supplies exactly those missing weights.

3.1 Fixing the multiple Gamma normalization

For an integer $r \geq 1$, the unit-period Barnes zeta is initially

$$\mathcal{Z}_r(u, x) = \sum_{n=0}^{\infty} \binom{n+r-1}{r-1} (n+x)^{-u}, \quad \Re u > r, \quad x > 0. \quad (3.1)$$

Define the explicit coefficient polynomials

$$C_{r,k}(x) = \frac{1}{(r-1)!} [X^k] \prod_{h=1}^{r-1} (X+h-x), \quad 0 \leq k < r. \quad (3.2)$$

Expanding the binomial coefficient in powers of $n+x$ gives

$$\mathcal{Z}_r(u, x) = \sum_{k=0}^{r-1} C_{r,k}(x) \zeta(u-k, x), \quad L_r(x) := \partial_u \mathcal{Z}_r(0, x) = \sum_{k=0}^{r-1} C_{r,k}(x) \zeta'(-k, x). \quad (3.3)$$

The first equality holds in the initial convergence domain and then meromorphically. The value at zero is regular. This is the classical Barnes/Hurwitz conversion, here written in a directly computable polynomial form [9, 10].

To specify a normalization completely, put

$$Q_r(x) = \sum_{j=0}^{r-1} (-1)^{j+1} L_{r-j}(1) \binom{x-1}{j}, \quad \log \Gamma_r(x) = L_r(x) + Q_r(x). \quad (3.4)$$

We use the division convention for the multiple Gamma hierarchy: $\Gamma_0(x) = x^{-1}$, $\Gamma_1 = \Gamma$, and $\Gamma_2 = G^{-1}$. The zeta prescription (3.3)–(3.4), not merely a functional equation, defines the normalized functions in every higher rank. Other standard sign conventions can be converted by inversion.

Proposition 3.1 (Normalization and recurrence). *The functions in (3.4) satisfy*

$$\Gamma_r(1) = 1, \quad \Gamma_r(x+1) = \frac{\Gamma_r(x)}{\Gamma_{r-1}(x)}. \quad (3.5)$$

Their normalization polynomials contain only rational numbers and ordinary zeta derivatives at nonpositive integers.

Proof. Counting the multiplicities in (3.1) gives $\mathcal{Z}_r(u, x+1) - \mathcal{Z}_r(u, x) = -\mathcal{Z}_{r-1}(u, x)$, with $\mathcal{Z}_0(u, x) = x^{-u}$. Hence $L_r(x+1) - L_r(x) = -L_{r-1}(x)$, where $L_0 = -\log x$. The binomial difference formula gives $Q_r(x+1) - Q_r(x) = -Q_{r-1}(x)$ for $r \geq 2$. For $r = 1$, $Q_1 = -\zeta'(0) = \frac{1}{2} \log(2\pi)$ is constant. Also $Q_r(1) = -L_r(1)$. These statements prove (3.5), including the usual Gamma recurrence at $r = 1$. The last claim follows from (3.3) at $x = 1$. $\square$

The first three rows make every coefficient explicit. Here $L = \log(2\pi)$:

$$L_1(x) = \zeta'(0, x), \quad Q_1(x) = L/2, \quad (3.6)$$

$$L_2(x) = \zeta'(-1, x) + (1-x)\zeta'(0, x), \quad Q_2(x) = -\zeta'(-1) - \frac{L}{2}(x-1), \quad (3.7)$$

$$L_3(x) = \frac{1}{2}\zeta'(-2, x) + \frac{3-2x}{2}\zeta'(-1, x) + \frac{(x-1)(x-2)}{2}\zeta'(0, x), \quad (3.8)$$

$$Q_3(x) = -\frac{1}{2}(\zeta'(-2) + \zeta'(-1)) + (x-1)\zeta'(-1) + \frac{L}{2}\binom{x-1}{2}. \quad (3.9)$$

At rank two these equations are equivalent to the usual Vardi identity

$$\log G(x) = \zeta'(-1) + (x-1)\log \Gamma(x) - \zeta'(-1, x). \quad (3.10)$$

The familiar Barnes function has its standard asymptotic normalization; $G(x+1) = \Gamma(x)G(x)$ and $G(1) = 1$ alone would allow multiplication by a nontrivial periodic factor.

3.2 The all-rank resolvent theorem

Theorem 3.2 (Individual multiple Gamma transforms). *For every integer $r \geq 1$, polynomial W , and $z \notin \mathbb{R}$,*

$$\boxed{\int_0^1 W(x) \log \Gamma_r(x) R_z(x) dx = - \sum_{k=0}^{r-1} \partial_s I_{WC_{r,k}}(s, z) \Big|_{s=k+1} + T_{WQ_r}(z).} \quad (3.11)$$

Every term on the right is explicitly given by (2.11) and (2.9). It is a finite combination of ordinary and depth-two polylogarithms, their first order derivatives, Gamma factors, logarithms, and ordinary zeta values and derivatives. All depth-two series needed at the stated evaluation points are normally convergent.

Proof. Differentiate the ordinary integral in Theorem 2.2 at $s = k + 1$:

$$\partial_s I_P(k+1, z) = - \int_0^1 P(x) \zeta'(-k, x) R_z(x) dx.$$

The shift formula at zero bounds the differentiated integrand by a fixed polynomial in $|\log x|$ times x when $k = 0$, and by an even more integrable power when $k > 0$; its analytic tail is bounded. Thus differentiation is justified locally uniformly. Insert (3.3) and (3.4). At every evaluation point $s = k + 1 \geq 1$ and every polynomial index $j \geq 1$, the depth-two total order is $s + j \geq 2$. Lemma 2.1 therefore applies with a margin, including all the stated spectral derivatives. The sums over r, k, j are finite. $\square$

This theorem answers the individual, unit-period multiple Gamma part of source question R6. It does not assert a reduction for arbitrary products of two multiple Gamma logarithms, nor for arbitrary incommensurable Barnes periods.

Repeated poles are equally explicit:

$$\int_0^1 \frac{W(x) \log \Gamma_r(x)}{(K(x) - z)^h} dx = \frac{1}{(h-1)!} \partial_z^{h-1} \int_0^1 W(x) \log \Gamma_r(x) R_z(x) dx, \quad h \geq 1. \quad (3.12)$$

All these integrals converge. Argument derivatives of $\mathcal{O}_{\alpha,\beta}(q)$ can be taken directly in its series, multiplying by powers of n . The formal identity

$$q\partial_q \mathcal{O}_{\alpha,\beta} = \mathcal{O}_{\alpha-1,\beta} - \mathcal{O}_{\alpha,\beta-1}$$

must be kept as a paired difference when either term separately leaves its convergence domain. The original differentiated series still converges normally. This qualification prevents an artificial divergence in implementations at total order two.

3.3 A compact formula for the ordinary Barnes function

Only one depth-two derivative is needed at rank two. Define

$$\mathcal{D}(q) = (\partial_\alpha + \partial_\beta) \mathcal{O}_{\alpha,\beta}(q)|_{\alpha=\beta=1} = - \sum_{n,m \geq 1} \frac{q^n \log(m(m+n))}{m(m+n)}. \quad (3.13)$$

It is holomorphic for $|q| < 1$ and vanishes at zero. In the next formulas all polylogarithms have argument q . Set

$$\mathcal{A}(q) = \frac{\mathcal{D}(q) - \text{Li}_2^{[1]}(q) - (\gamma + L + 1) \text{Li}_2(q) - \text{Li}_1(q) \text{Li}_1^{[1]}(q)}{2\pi^2} - \frac{1}{4} \text{Li}_1(q), \quad (3.14)$$

$$\mathcal{S}(q) = \frac{\text{Li}_1^{[1]}(q) - (\gamma + 2L) \text{Li}_1(q) - \frac{1}{2}(\text{Li}_2(q) + \text{Li}_1(q)^2)}{2\pi}, \quad (3.15)$$

$$\mu_G = 2\zeta'(-1) - \frac{1}{12} - \frac{L}{4}. \quad (3.16)$$

Theorem 3.3 (The individual G resolvent). *With the coordinates (2.1),*

$$\boxed{\int_0^1 \frac{\log G(x)}{\pi \cot(\pi x) - z} dx = -\frac{\mu_G}{a} + \frac{c}{2q} (\mathcal{A}(q) + \epsilon i \mathcal{S}(q))}. \quad (3.17)$$

The quotient at $q = 0$ is removable. The formula is holomorphic on each open half-plane and uses the real logarithm of $G(x)$ on the integration interval.

Proof. We give a summation proof that also connects the formula to the known Barnes Fourier expansion [11, Theorems 2–3]. Write

$$\log G(x) = \mu_G + \sum_{n \geq 1} (a_n \cos(2\pi n x) + b_n \sin(2\pi n x)).$$

Its coefficients are

$$a_n = \frac{\frac{1}{2} \log n - (\gamma + L) - 1}{2\pi^2 n^2} - \frac{1}{4n} - \frac{T_n}{\pi^2}, \quad (3.18)$$

$$b_n = \frac{\frac{1}{2n} - \gamma - 2L - \log n - H_n}{2\pi n}, \quad (3.19)$$

$$T_n = \sum_{\substack{m \geq 1 \\ m \neq n}} \frac{\log m}{m^2 - n^2}. \quad (3.20)$$

The m -sum in T_n converges absolutely. These coefficients can also be recovered from Theorem 2.2 by differentiating the $P = x - 1$ formula at $s = 1$, the $P = 1$ formula at $s = 2$, and using (3.10). Thus their use does not assume a new Barnes Fourier theorem.

To evaluate their generating functions, differentiate the $j = 1$ case of (2.13) and the definition of $\mathcal{O}_{1,s}$. Partial fractions give the exact identity

$$\sum_{n \geq 1} q^n T_n = \frac{1}{2} \text{Li}_1(q) \text{Li}_1^{[1]}(q) - \frac{1}{2} \mathcal{D}(q) + \frac{1}{4} \text{Li}_2^{[1]}(q). \quad (3.21)$$

For clarity, the scalar calculation behind it is

$$\frac{1}{m^2 - n^2} = \frac{1}{2m} \left(\frac{1}{m - n} + \frac{1}{m + n} \right).$$

The excluded term $m = n$ in the second sum is $\log n / (2n^2)$ before multiplication by $1/2$; it supplies the last term in (3.21). This diagonal term is necessary for the coefficient of $\text{Li}_2^{[1]}$.

Next,

$$\sum_{n \geq 1} \frac{H_n q^n}{n} = \text{Li}_2(q) + \frac{1}{2} \text{Li}_1(q)^2. \quad (3.22)$$

It follows either by integrating $\sum_{n \geq 1} H_n q^n = \text{Li}_1(q)/(1 - q)$ or by the depth-two stuffle identity. Equations (3.18)–(3.22) give $\sum a_n q^n = \mathcal{A}(q)$ and $\sum b_n q^n = \mathcal{S}(q)$.

The function $\log G$ is integrable and square integrable: near zero it equals $\log x + O(x)$, while it is analytic at one. Pair its Fourier coefficients with the uniformly convergent geometric series (2.2). This yields (3.17). The constant term is (3.16); for example integrate (3.10), use $\int_0^1 \zeta'(-1, x) dx = 0$ and

$$\int_0^1 x \log \Gamma(x) dx = \frac{L}{4} + \zeta'(-1) - \frac{1}{12}.$$

The latter follows by integrating the Hurwitz argument ladder. All generating functions used above are $O(q)$, so the quotient at zero is removable. Normal convergence proves holomorphy. $\square$

3.4 An ordinary-polylogarithm real component

Corollary 3.4 (A one-parameter exact real integral). *For $b > 0$, let $q = (b - \pi)/(b + \pi) \in (-1, 1)$. Then*

$$\boxed{\int_0^1 \frac{K(x) \log G(x)}{K(x)^2 + b^2} dx = \frac{\text{Li}_1^{[1]}(q) - (\gamma + 2L) \text{Li}_1(q) - \frac{1}{2}(\text{Li}_2(q) + \text{Li}_1(q)^2)}{2q(b + \pi)^2}.} \quad (3.23)$$

At $b = \pi$ the right side means $-(\gamma + 2L + 1/2)/(8\pi^2)$. In particular,

$$\boxed{\int_0^1 \frac{\pi \cot(\pi x) \log G(x)}{\pi^2 \cot^2(\pi x) + 9\pi^2} dx = \frac{\text{Li}_1^{[1]}(1/2) - (\gamma + 2 \log(2\pi)) \log 2 - \pi^2/24 - (\log 2)^2/4}{16\pi^2}.} \quad (3.24)$$

Proof. At $z = ib$, $a = i(b + \pi)$ and $c = -2\pi i/(b + \pi)^2$. The functions $\mathcal{A}, \mathcal{S}$ are real at real q . Taking the real part of (3.17) therefore keeps only $\pi \mathcal{S}(q)/(q(b + \pi)^2)$. Use $\text{Li}_1(q)/q \rightarrow 1$, $\text{Li}_2(q)/q \rightarrow 1$ and $\text{Li}_1^{[1]}(q)/q \rightarrow 0$ at zero. For $b = 3\pi$, use $q = 1/2$, $\text{Li}_1(1/2) = \log 2$ and $\text{Li}_2(1/2) = \pi^2/12 - (\log 2)^2/2$. $\square$

The integral (3.24) is approximately $-0.02312768479852353748486907193849$. The complete complex transform at $z = 3\pi i$ is approximately

$$-0.02312768479852353748486907193849 - 0.05510836371472001971368271205301 i.$$

These decimals are diagnostics, not definitions or exact-value certificates. The symmetry of K under $x \mapsto 1 - x$ explains why its real component isolates the simpler sine coefficients.

3.5 Normalized antiderivatives remain explicit

The standard balanced primitives are

$$\mathcal{B}_k(x) = \frac{\zeta'(-k, x) + H_k \zeta(-k, x)}{k!}, \quad H_0 = 0. \quad (3.25)$$

They satisfy $\mathcal{B}_0 = \log \Gamma - L/2$ and $\mathcal{B}'_k = \mathcal{B}_{k-1}$ for $k \geq 1$ [7, 8]. For a polynomial P , set

$$\mathcal{A}_{P,k}(x) = \sum_{j=0}^{\deg P} (-1)^j P^{(j)}(x) \mathcal{B}_{k+j+1}(x). \quad (3.26)$$

Termwise differentiation telescopes, so $\mathcal{A}'_{P,k} = P \mathcal{B}_k$. Consequently a primitive of $P(x) \zeta'(-k, x)$, normalized to vanish at $x = 1$, is

$$k!(\mathcal{A}_{P,k}(x) - \mathcal{A}_{P,k}(1)) - H_k \int_1^x P(t) \zeta(-k, t) dt. \quad (3.27)$$

The remaining integral is polynomial because $\zeta(-k, t) = -B_{k+1}(t)/(k+1)$. Apply (3.27) to the finitely many terms in (3.3), and integrate Q_r as a polynomial. This gives a fully normalized primitive of every $\log \Gamma_r$. Repetition gives every iterated primitive, without unspecified additive constants. Their resolvent transforms again follow from Theorem 3.2 and Theorem 2.2. The primitive operation is classical integration by parts; the point here is its compatibility with the explicit individual-function normalization.

4 Complex cotangent powers and their resonant amplitudes

Question R4 of the incoming *Harmonic Parity and Resolvent Identities* report [2] asks for a joint generator for nonintegral powers of the cotangent resolvent, with the moving endpoint amplitudes retained before expansion. We prove such a generator and its finite-part conversion. The primary result below covers every complex exponent with positive real part and every positive integral crossing. A new Dirichlet-series coordinate is used explicitly; no finite reduction to ordinary polylogarithms is asserted for an arbitrary complex exponent. Nevertheless, its exponent jets have a finite multiple-polylogarithmic description, and one special resolvent point has a Gamma evaluation at every positive integral spectral order.

4.1 Branches and a binomial Dirichlet transform

Put

$$K(x) = \pi \cot(\pi x), \quad R_z(x) = (K(x) - z)^{-1}, \quad z \in \mathbb{C} \setminus \mathbb{R}.$$

For the chosen half-plane define

$$\epsilon = \operatorname{sgn} \Im z, \quad a = z + \epsilon \pi i, \quad A = -a^{-1}, \quad q = \frac{z - \epsilon \pi i}{z + \epsilon \pi i}, \quad \sigma = -2\epsilon \pi i.$$

Then $|q| < 1$. We take the continuous logarithm of $K(x) - z$ whose imaginary part lies in $(-\pi, 0)$ if $\epsilon = 1$ and in $(0, \pi)$ if $\epsilon = -1$. Consequently

$$R_z(x)^\lambda = \exp\{-\lambda \operatorname{Log}(K(x) - z)\}.$$

Also $\operatorname{Log} \sigma = \log(2\pi) - \epsilon \pi i/2$, and $\operatorname{Log} A$ is the principal logarithm.

With $w = e^{2\epsilon \pi i x}$, elementary algebra gives

$$R_z(x)^\lambda = A^\lambda (1 - w)^\lambda (1 - qw)^{-\lambda}. \quad (4.1)$$

Here the two logarithms on the right are their holomorphic determinations in $|w| < 1$, continued to $w \neq 1$ on its boundary. To check the constant in this identity, both sides divided by x^λ tend to one as $x \downarrow 0$. This fixes the phase at the other endpoint as well.

Define

$$\begin{aligned} h_n(\lambda, q) &= [w^n] (1 - w)^\lambda (1 - qw)^{-\lambda} \\ &= \sum_{k=0}^n \frac{(-\lambda)_k (\lambda)_{n-k}}{k! (n-k)!} q^{n-k}, \end{aligned} \quad (4.2)$$

$$\mathcal{L}_\lambda(u, q) = \sum_{n=1}^{\infty} \frac{h_n(\lambda, q)}{n^u}. \quad (4.3)$$

The Pochhammer symbols have their polynomial meaning, including at integral parameters. The initial Dirichlet series converges normally where $\Re(u + \lambda) > 0$, locally in $|q| < 1$. One way to see the needed coefficient estimate is to convolve the binomial coefficients of $(1 - w)^\lambda$ with the exponentially decaying coefficients of $(1 - qw)^{-\lambda}$; on compact parameter sets this gives $h_n = O(n^{-\Re \lambda - 1})$, with a harmless uniform weakening of the exponent at the boundary of a compact set.

Theorem 4.1 (Joint continuation of the binomial transform). *The function $\mathcal{L}_\lambda(u, q)$ continues meromorphically in $(u, \lambda) \in \mathbb{C}^2$, holomorphically in $|q| < 1$. Its possible polar hypersurfaces are*

$$u + \lambda = -j, \quad j = 0, 1, 2, \dots$$

Away from them,

$$\mathcal{L}_\lambda(0, q) = -1, \quad \mathcal{L}_\lambda(-N, q) = 0 \quad (N \geq 1).$$

These equalities concern the meromorphic restriction at the indicated spectral order. At an intersection with a polar hypersurface, substitution and continuation need not commute.

Proof. For $\Re u > 0$ and $\Re(u + \lambda) > 0$, termwise integration gives

$$\Gamma(u) \mathcal{L}_\lambda(u, q) = \int_0^\infty t^{u-1} \left[\left(\frac{1 - e^{-t}}{1 - qe^{-t}} \right)^\lambda - 1 \right] dt. \quad (4.4)$$

The positive real logarithm of $1 - e^{-t}$, and the holomorphic logarithm of $1 - qe^{-t}$, specify its powers. Set

$$\left(\frac{1 - e^{-t}}{1 - qe^{-t}} \right)^\lambda = t^\lambda \sum_{j \geq 0} b_j(\lambda, q) t^j.$$

The coefficients are entire in λ , holomorphic in q . For any fixed N , split the integral at one and subtract the first N terms of this local expansion. The result is

$$\Gamma(u)\mathcal{L}_\lambda(u, q) = -\frac{1}{u} + \sum_{j=0}^{N-1} \frac{b_j(\lambda, q)}{u + \lambda + j} + E_N(u, \lambda, q), \quad (4.5)$$

where E_N is holomorphic when $\Re(u + \lambda) > -N$. The tail at infinity is exponentially convergent on every compact parameter set. Increasing N proves the continuation. Division by $\Gamma(u)$ removes the isolated pole $-1/u$, with value -1 at $u = 0$, and gives zero at the other nonpositive integers whenever no moving denominator vanishes. $\square$

4.2 The completed spectral generator

Use the conventional normalization

$$\zeta(1-s, x) = -\frac{1}{s} + \sum_{m \geq 0} \gamma_m(x) \frac{s^m}{m!}, \quad \gamma_0(x) = -\psi(x).$$

For an integer $p \geq 0$, let

$$A_p(s) = \prod_{k=1}^p (s - k), \quad A_0(s) = 1.$$

Argument differentiation satisfies

$$\partial_x^p \zeta(1-s, x) = A_p(s) \zeta(1+p-s, x).$$

The notation $\gamma_m^{(p)}$ below always means argument differentiation.

Theorem 4.2 (Complex-power cotangent generator). *In the ordinary-integral region $\Re \lambda > -1$, $\Re(s + \lambda) > p$, away from the Hurwitz pole when $p = 0$,*

$$\mathcal{J}_p(s, \lambda, z) := \int_0^1 \partial_x^p \zeta(1-s, x) R_z(x)^\lambda dx = A^\lambda \Gamma(s) \sigma^{p-s} \mathcal{L}_\lambda(s-p, q). \quad (4.6)$$

This identity defines a joint meromorphic continuation to all (s, λ) . Besides the pole $-A^\lambda/s$ when $p = 0$, its only possible polar hypersurfaces are

$$s + \lambda = p - j, \quad j = 0, 1, \dots$$

Define the completed family

$$\mathcal{F}_p(s, \lambda, z) = \mathcal{J}_p(s, \lambda, z) + \mathbf{1}_{p=0} \frac{A^\lambda}{s}. \quad (4.7)$$

For $\Re \lambda > p$ the ordinary absolutely convergent moments

$$\mathcal{M}_{m,p}(\lambda, z) = \int_0^1 \gamma_m^{(p)}(x) R_z(x)^\lambda dx = m! [s^m] \mathcal{F}_p(s, \lambda, z) \quad (4.8)$$

therefore have meromorphic continuations in λ , with possible poles only at the integers $\lambda \leq p$, of order at most $m + 1$.

Proof. First suppose $\Re \lambda > 0$ and $\Re s > p + 1$. The Fourier series of the kernel in (4.1) is absolutely convergent. The classical Hurwitz Fourier identity [4, 25.11.9] gives

$$\int_0^1 \zeta(1-s, x) dx = 0, \quad \int_0^1 \partial_x^p \zeta(1-s, x) e^{2\pi i n x} dx = \Gamma(s) \sigma^{p-s} n^{p-s}.$$

The constant kernel mode vanishes, and summing the other modes proves (4.6). The endpoint decomposition

$$\partial_x^p \zeta(1-s, x) = A_p(s) x^{s-p-1} + \partial_x^p \zeta(1-s, 1+x)$$

shows normal integrability on the larger region stated in the theorem, so the identity extends there.

In (4.5), put $u = s - p$ and use $\Gamma(s)/\Gamma(s-p) = A_p(s)$. The moving denominators give exactly the indicated hypersurfaces. The isolated term $-1/(s-p)$ is cancelled by $A_p(s)$ for $p \geq 1$; for $p = 0$ it gives residue $-A^\lambda$. This proves the claimed completion. Finally, the mean of the kernel is A^λ for $\Re \lambda > 0$, by its constant Fourier coefficient. Termwise Stieltjes expansion is valid when $\Re \lambda > p$, including all powers of the endpoint logarithm. This proves (4.8) and its continuation. $\square$

Corollary 4.3 (Lowest Stieltjes index and shift identities). *Away from the moving poles,*

$$\begin{aligned} \mathcal{M}_{0,0}(\lambda, z) &= A^\lambda (\gamma + \text{Log } \sigma + \partial_u \mathcal{L}_\lambda(u, q)|_{u=0}), \\ \mathcal{M}_{0,p}(\lambda, z) &= A^\lambda \sigma^p \partial_u \mathcal{L}_\lambda(u, q)|_{u=-p} \quad (p \geq 1). \end{aligned}$$

The continued moments satisfy the meromorphic identities

$$\partial_z \mathcal{M}_{m,p}(\lambda, z) = \lambda \mathcal{M}_{m,p}(\lambda + 1, z), \quad (4.9)$$

$$\begin{aligned} \mathcal{M}_{m,p+1}(\lambda, z) &= -\lambda (\mathcal{M}_{m,p}(\lambda - 1, z) + 2z \mathcal{M}_{m,p}(\lambda, z) \\ &\quad + (z^2 + \pi^2) \mathcal{M}_{m,p}(\lambda + 1, z)). \end{aligned} \quad (4.10)$$

Proof. The first two formulas follow by expanding $\Gamma(s)\sigma^{-s}$ and using Theorem 4.1 at $0, -p$. The first shift follows from $\partial_z R_z^\lambda = \lambda R_z^{\lambda+1}$. For the second, integrate by parts where $\Re \lambda > p + 1$, so the boundary terms vanish, and use

$$R'_z = 1 + 2z R_z + (z^2 + \pi^2) R_z^2.$$

Continuation proves the meromorphic statements. At a resonant integer, a pointwise finite-part version must use the conversion in the next theorem before substituting. $\square$

4.3 All positive resonances and the two finite parts

Finite parts in the next theorem use the original x -coordinate: they retain the constant coefficient in the cutoff integral $\int_\eta^{1-\eta}$, after removing every nonintegrable power and logarithm. For a complex exponent of positive real part this definition is unambiguous from the local power-log expansion. It agrees with the ordinary integral whenever that integral converges.

The complete local amplitude is elementary. Write

$$D_j(\lambda, z) = [x^j] (\pi x \cot(\pi x) - zx)^{-\lambda}. \quad (4.11)$$

Thus $R_z(x)^\lambda = x^\lambda \sum_{j \geq 0} D_j(\lambda, z) x^j$ near zero, with

$$D_0 = 1, \quad D_1 = \lambda z, \quad D_2 = \frac{\lambda \pi^2}{3} + \frac{\lambda(\lambda+1)z^2}{2}.$$

Every D_j is a polynomial in λ, z, π^2 with rational coefficients, and has degree at most j in λ .

Theorem 4.4 (Resonant amplitudes and finite-part conversion). *For a positive integer $r \leq p$, put $j = p - r$. In a neighborhood of $(s, \lambda) = (0, r)$,*

$$\mathcal{F}_p(s, \lambda, z) = \frac{A_p(s)D_j(\lambda, z)}{s + \lambda - r} + H_{p,r}(s, \lambda, z), \quad (4.12)$$

where $H_{p,r}$ is jointly holomorphic. Hence, for every positive integer r ,

$$\begin{aligned} & \text{FP}_x \int_0^1 \gamma_m^{(p)}(x) R_z(x)^r dx \\ &= m! [s^m] \left\{ \mathcal{F}_p(s, r, z) - \mathbf{1}_{r \leq p} \frac{A_p(s)D_{p-r}(r, z)}{s} \right\}. \end{aligned} \quad (4.13)$$

There is also an explicit comparison with continuation in the kernel exponent. Write $A_p(s) = \sum_{h=0}^p a_{p,h} s^h$. For $1 \leq r \leq p$, define

$$\Delta_{m,p,r}(z) = m! \sum_{h=0}^{\min(p,m)} \frac{(-1)^{m-h} a_{p,h}}{(m-h+1)!} \partial_\lambda^{m-h+1} D_{p-r}(\lambda, z) \Big|_{\lambda=r}. \quad (4.14)$$

Then

$$\text{FP}_x \int_0^1 \gamma_m^{(p)}(x) R_z(x)^r dx = \text{CT}_{\lambda=r} \mathcal{M}_{m,p}(\lambda, z) - \Delta_{m,p,r}(z). \quad (4.15)$$

For $\Re \lambda > 0$ away from the positive integers $1, \dots, p$, the meromorphic value of $\mathcal{M}_{m,p}$ equals the pointwise x -finite part.

Proof. Subtract a sufficiently long full local Taylor jet, so that the remaining endpoint integral converges normally near $(0, r)$. Among the explicitly integrated Taylor terms, the only one whose exponent crosses -1 there is

$$A_p(s)D_{p-r}(\lambda, z)x^{s+\lambda-r-1}.$$

Use a smooth cutoff on a small interval next to zero. The continued integrals of all other Taylor terms, and the normally convergent remainder, are holomorphic near $(0, r)$. The distinguished term has meromorphic integral with polar part $A_p(s)D_{p-r}(\lambda, z)/(s + \lambda - r)$. Replacing the cutoff by one only changes a holomorphic function; this proves (4.12) with the indicated full amplitude, not just its value on the polar hyperplane.

At fixed $\lambda = r$, the analytically continued integral of the resonant local term is $A_p(s)D_{p-r}(r, z)/s$. Every coefficient of x^{s-1} is a power of $\log x$ divided by x , whose finite part on $(0, 1)$ is zero. Deleting the entire amplitude before expansion proves (4.13). Terms on the cutoff complement cancel the artificial splitting point.

For the second formula, put $\delta = \lambda - r$ and take the s^m coefficient of the polar term in (4.12). It is

$$m! \sum_{h=0}^{\min(p,m)} (-1)^{m-h} a_{p,h} \frac{D_{p-r}(r + \delta, z)}{\delta^{m-h+1}}.$$

Its constant Laurent coefficient is exactly (4.14). The holomorphic remainder contributes the same value along either order of expansion. This proves (4.15). Away from a crossing, endpoint subtractions have nonzero exponents, and coefficient extraction commutes with finite-part integration. $\square$

The theorem supplies the full principal part as well: expand the polynomial $D_{p-r}(r + \delta, z)$ in the last display and retain negative powers of δ . Some potential poles cancel. In particular, $\Delta_{m,p,p} = 0$, since $D_0 = 1$; a collision does not automatically imply a nonzero conversion term.

A nonzero conversion for the simple resolvent. The simplest example is $m = 0, p = 2, r = 1$. Since $A_2(s) = s^2 - 3s + 2$ and $D_1(\lambda, z) = \lambda z$,

$$\Delta_{0,2,1}(z) = 2z.$$

At $z = i\pi$, set $\ell = \log(2\pi) - i\pi/2$. The finite Fourier series $R_{i\pi}(x) = (e^{2\pi i x} - 1)/(2\pi i)$ gives

$$\text{FP}_x \int_0^1 \gamma_0''(x) R_{i\pi}(x) dx = 2\pi i \left(\frac{3}{2} - \gamma - \ell \right).$$

It follows that

$$\text{CT}_{\lambda=1} \mathcal{M}_{0,2}(\lambda, i\pi) = 2\pi i \left(\frac{5}{2} - \gamma - \ell \right). \quad (4.16)$$

The two constants differ by $2\pi i$. This is a concrete obstruction to identifying exponent continuation with the specified x -finite part. In contrast the case $m = 0, p = r = 1$ has no exponent-axis correction.

Exact recovery of integral powers. For a positive integer r and $q \neq 0$, partial fractions give

$$h_n(r, q) = q^{n-r} \sum_{j=1}^r \binom{r}{j} (q-1)^j \binom{n+j-1}{j-1} \quad (n \geq 1).$$

The binomial on the right is a polynomial of degree $j-1$ in n . Consequently $\mathcal{L}_r(u, q)$ is a finite linear combination of the ordinary $\text{Li}_{u-k}(q)$, $0 \leq k < r$, with rational coefficients in q . At $q = 0$ the removable limiting expression is

$$\mathcal{L}_r(u, 0) = \sum_{n=1}^r \frac{(-1)^n \binom{r}{n}}{n^u}.$$

Thus the completed generator recovers precisely the ordinary polylogarithm class of the previous repeated-resolvent theorem.

4.4 Gamma closure at a special point

For $z = \epsilon i\pi$ one has $q = 0$, and

$$\mathcal{L}_\lambda(u, 0) = \sum_{n \geq 1} \frac{(-\lambda)_n}{n! n^u}.$$

At every positive integer u , this new coordinate reduces to generalized harmonic numbers at a complex argument. Define

$$H_\lambda^{(1)} = \psi(1 + \lambda) + \gamma, \quad H_\lambda^{(r)} = \zeta(r) - \zeta(r, 1 + \lambda) \quad (r \geq 2),$$

initially for $\Re \lambda > -1$ and then meromorphically.

Theorem 4.5 (Every integral spectral order at $q = 0$). *For $N \geq 1$,*

$$\mathcal{L}_\lambda(N, 0) = -\frac{1}{N!} Y_N(H_\lambda^{(1)}, 1! H_\lambda^{(2)}, \dots, (N-1)! H_\lambda^{(N)}), \quad (4.17)$$

where Y_N is the complete exponential Bell polynomial. Equivalently, near $v = 0$,

$$\sum_{N \geq 1} \mathcal{L}_\lambda(N, 0) v^{N-1} = \frac{1}{v} \left\{ 1 - \frac{\Gamma(1-v)\Gamma(1+\lambda)}{\Gamma(1+\lambda-v)} \right\}. \quad (4.18)$$

In particular,

$$\mathcal{L}_\lambda(1, 0) = -H_\lambda^{(1)}, \quad \mathcal{L}_\lambda(2, 0) = -\frac{(H_\lambda^{(1)})^2 + H_\lambda^{(2)}}{2}.$$

Proof. For $\Re \lambda > -1$ and $\Re t > 0$, the beta integral gives

$$\sum_{n \geq 1} \frac{(-\lambda)_n}{n!(n+t)} = \int_0^1 y^{t-1} ((1-y)^\lambda - 1) dy = \frac{\Gamma(t)\Gamma(1+\lambda)}{\Gamma(1+\lambda+t)} - \frac{1}{t}.$$

Expand at $t = -v$. The left side generates the positive integral spectral orders. On the right,

$$\log \frac{\Gamma(1-v)\Gamma(1+\lambda)}{\Gamma(1+\lambda-v)} = \sum_{r \geq 1} \frac{H_\lambda^{(r)}}{r} v^r.$$

The defining Bell-polynomial generating function proves both identities. Meromorphic continuation removes the initial restrictions. This is a beta-integral specialization, not a claim of priority for the classical binomial-harmonic identity. $\square$

4.5 Exponent jets and colored multiple polylogarithms

There is also an exact finite description of every exponent jet at $\lambda = 0$. For $q \neq 0$, $|q| < 1$, let $\mathcal{A} = \{1, q\}$, with signs $\nu(1) = -1, \nu(q) = 1$. The multiple polylogarithm uses decreasing indices:

$$\text{Li}_{s_1, \dots, s_d}(z_1, \dots, z_d) = \sum_{n_1 > \dots > n_d \geq 1} \frac{z_1^{n_1} \dots z_d^{n_d}}{n_1^{s_1} \dots n_d^{s_d}}.$$

Proposition 4.6 (All exponent jets at zero). *For $d \geq 1$ and $\Re u > 0$,*

$$\partial_\lambda^d \mathcal{L}_\lambda(u, q) \Big|_{\lambda=0} = d! \sum_{a_1, \dots, a_d \in \mathcal{A}} \left(\prod_{j=1}^d \nu(a_j) \right) \text{Li}_{u+1, \{1\}^{d-1}} \left(a_1, \frac{a_2}{a_1}, \dots, \frac{a_d}{a_{d-1}} \right). \quad (4.19)$$

All displayed nested sums converge absolutely. Their cumulative color products are $a_j \in \{1, q\}$, so the appearance of q^{-1} as an individual color causes no convergence problem. For nonintegral u , these are nested Dirichlet sums with a complex first index; at positive integral u , they are classical colored multiple polylogarithms. At $q = 0$, the continuous limiting identity is

$$\partial_\lambda^d \mathcal{L}_\lambda(u, 0) \Big|_{\lambda=0} = (-1)^d d! \zeta(u+1, \{1\}^{d-1}).$$

Proof. Put $F(w) = \text{Li}_1(qw) - \text{Li}_1(w)$. The d -th exponent derivative of the disk kernel at zero is $F(w)^d$. Repeated integration of

$$F'(w) = \frac{q}{1-qw} - \frac{1}{1-w}$$

proves the shuffle identity

$$\frac{F(w)^d}{d!} = \sum_{a_1, \dots, a_d \in \mathcal{A}} \left(\prod_j \nu(a_j) \right) \text{Li}_{\{1\}^d} \left(a_1 w, \frac{a_2}{a_1}, \dots, \frac{a_d}{a_{d-1}} \right).$$

Multiplying the coefficient of w^n by n^{-u} raises the first index from one to $u+1$. Absolute convergence for $\Re u > 0$ justifies evaluation at $w = 1$ and all coefficient interchanges. If $q = 0$, only the latter one survives. $\square$

The first jet is the simple identity

$$\partial_\lambda \mathcal{L}_\lambda(u, q)|_0 = \text{Li}_{u+1}(q) - \zeta(u+1).$$

Higher jets explain how generalized harmonic sums and increasing multiple-polylogarithm depth enter complex powers. The finite description is proved for each fixed jet; it does not imply finite ordinary-polylogarithm closure after summing all jets.

Verification. The accompanying program independently compares the spectral generator with ordinary x -quadrature at nonintegral complex parameters, checks the Gamma specialization against Mellin quadrature, and compares a nearby-exponent Laurent extrapolation with (4.16). Its recorded numerical residuals are diagnostics. The continuation, pole classification, and all-order conversion rest on the local subtractions above.

5 The fourth digamma moment and harmonic Laurent data

The incoming *Cubic Tornheim Derivatives and Harmonic Stieltjes Constants* report, Section 7, Question 6 [3], explicitly asks for a power-to-harmonic formula at the fourth power. The following theorem answers that question. It is separate from the quartic *directional* Tornheim layer, which has already been worked out in later incoming reports. No claim of worldwide novelty or finite reduction to ordinary zeta constants is intended.

Write $H_n^{(r)} = \sum_{j=1}^n j^{-r}$, $H_n = H_n^{(1)}$, and

$$H_n(1, 1) = \sum_{n \geq j > k \geq 1} \frac{1}{jk} = \frac{H_n^2 - H_n^{(2)}}{2}.$$

The strict harmonic Dirichlet series used here are

$$Z_2(s, 1) = \sum_{n=1}^{\infty} \frac{H_{n-1}}{n^s}, \quad Z_3(s, 1, 1) = \sum_{n=1}^{\infty} \frac{H_{n-1}(1, 1)}{n^s}, \quad \Re s > 1.$$

They have the local expansions

$$Z_2(1+u, 1) = u^{-2} + \gamma u^{-1} + \frac{\gamma^2 - \zeta(2)}{2} + \eta u + O(u^2), \quad (5.1)$$

$$Z_3(1+u, 1, 1) = u^{-3} + \gamma u^{-2} + \frac{\gamma^2 - \zeta(2)}{2u} + \frac{\gamma^3 - 3\gamma\zeta(2) + 2\zeta(3)}{6} + \beta_1 u + O(u^2). \quad (5.2)$$

In particular β_1 is a completely specified strict depth-three harmonic Laurent coefficient; it is not a coefficient of the inclusive harmonic series. The constant term is also fixed by the Mellin lemma below. The pole terms in the second expansion follow directly from $H_{n-1} = \log n + \gamma + O(n^{-1})$ and $H_{n-1}^{(2)} = \zeta(2) + O(n^{-1})$, by subtraction of zeta derivatives. The convention for the ordinary Stieltjes constants is

$$\zeta(1+u) = u^{-1} + \sum_{j \geq 0} \frac{(-1)^j \gamma_j}{j!} u^j.$$

Theorem 5.1 (A quartic power-to-harmonic identity). *Use the unit-coordinate Hadamard finite part at the origin and put $Q_4 = \text{FP} \int_0^1 \psi(x)^4 dx$. Then*

$$Q_4 = 18\zeta(4) + 12\zeta(3) - 12\gamma\zeta(2) + 4\zeta'(3) + 12(\gamma^2 - \zeta(2))\gamma_1 + 24\gamma\eta - 24\beta_1. \quad (5.3)$$

An equivalent absolutely convergent arithmetic formula is

$$Q_4 = 18\zeta(4) + 12\zeta(3) - 12\gamma\zeta(2) + 4\zeta'(3) + 12\gamma_3 - 24\zeta(2)\gamma_1 + 12 \sum_{n=1}^{\infty} \frac{(\psi(n)^2 + \psi'(n) - \log^2 n) \log n}{n}. \quad (5.4)$$

If $Q_3 = \text{FP} \int_0^1 \psi(x)^3 dx$, the already proved cubic bridge [3] $Q_3 = 3\zeta(2) - 6\eta - 6\gamma\gamma_1$ eliminates η :

$$Q_4 + 4\gamma Q_3 = 18\zeta(4) + 12\zeta(3) + 4\zeta'(3) - 12(\gamma^2 + \zeta(2))\gamma_1 - 24\beta_1. \quad (5.5)$$

5.1 A normalized Mittag–Leffler expansion

For $n \geq 0$ set

$$\begin{aligned} b_n &= H_n - \gamma, & c_n &= \zeta(2) + H_n^{(2)}, & e_n &= H_n^{(3)} - \zeta(3), \\ a_n &= c_n - b_n^2, & d_n &= 6b_n^2 - 4c_n, & r_n &= -4b_n^3 + 12b_n c_n - 4e_n. \end{aligned}$$

Let

$$C_* = \gamma^4 - 12\gamma^2\zeta(2) + 12\gamma\zeta(3) + 11\zeta(4), \quad P_n(z) = \frac{1}{(n+z)^4} - \frac{4b_n}{(n+z)^3} + \frac{d_n}{(n+z)^2} + \frac{r_n}{n+z}.$$

Lemma 5.2 (Quartic principal parts with no unknown entire term). *For $z \notin \{0, -1, -2, \dots\}$,*

$$\psi(z)^4 = P_0(z) + C_* + \sum_{n=1}^{\infty} (P_n(z) - P_n(0)). \quad (5.6)$$

The series converges normally on every compact set avoiding the poles.

Proof. The digamma recurrence and its convergent expansion at zero [5] give

$$\psi(-n + \varepsilon) = -\varepsilon^{-1} + b_n + c_n\varepsilon + e_n\varepsilon^2 + (\zeta(4) + H_n^{(4)})\varepsilon^3 + O(\varepsilon^4).$$

Raising this expression to the fourth power gives exactly P_n as its principal part. The constant coefficient at $n = 0$ is

$$\gamma^4 - 12\gamma^2\zeta(2) + 6\zeta(2)^2 + 12\gamma\zeta(3) - 4\zeta(4) = C_*.$$

This establishes the poles but does not on its own establish the entire remainder. To determine that remainder, let $f(z) = \psi(z)^4 - P_0(z)$. For fixed z , apply the residue theorem to $zf(w)/(w(w-z))$ on $|w| = N + \frac{1}{2}$. The standard digamma asymptotic in the right half-plane, and $\psi(w) = \psi(1-w) - \pi \cot(\pi w)$ in the left half-plane, give $f(w) = O(\log^4 N)$ uniformly on these circles. The cotangent is uniformly bounded there: away from the real axis this follows from its exponential formula, while near the real axis these half-integer circles remain a fixed positive distance from the integer poles. Thus the contour integral is $O(\log^4 N/N)$ and tends to zero. The residues are $f(z)$, $-f(0)$, and $-(P_n(z) - P_n(0))$, proving the formula. Finally $b_n = O(\log n)$, $d_n = O(\log^2 n)$ and $r_n = O(\log^3 n)$, and the simple-pole difference is $O(n^{-2})$ on each fixed compact set. These estimates prove normal convergence. $\square$

5.2 The exact summation-by-parts cancellation

For $m \geq 2$ and $n \geq 1$,

$$\int_0^1 ((n+x)^{-m} - n^{-m}) dx = \frac{n^{1-m} - (n+1)^{1-m}}{m-1} - n^{-m}.$$

Integrating (5.6), and retaining the finite parts of its three higher-order poles at zero, yields

$$Q_4 = K + \sum_{n=1}^{\infty} r_n \left(\log \left(1 + \frac{1}{n} \right) - \frac{1}{n} \right), \quad (5.7)$$

where

$$K = \gamma^4 - 18\gamma^2\zeta(2) + 32\gamma\zeta(3) + \frac{13}{2}\zeta(4) + 12\zeta(3) - 12\gamma\zeta(2). \quad (5.8)$$

For clarity, the ordinary Euler sums needed in this simplification are

$$\sum_{n \geq 1} \frac{H_n}{n^2} = 2\zeta(3), \quad \sum_{n \geq 1} \frac{H_n}{n^3} = \frac{5}{4}\zeta(4), \quad \sum_{n \geq 1} \frac{H_n^{(2)}}{n^2} = \frac{7}{4}\zeta(4), \quad \sum_{n \geq 1} \frac{H_n^2}{n^2} = \frac{17}{4}\zeta(4). \quad (5.9)$$

These are classical ordinary convergent sums. One short derivation of the three weight-four formulas is as follows. The shuffle and stuffle products of $\zeta(2)^2$ give respectively $2\zeta(2,2) + 4\zeta(3,1)$ and $2\zeta(2,2) + \zeta(4)$; hence $\zeta(3,1) = \zeta(4)/4$ and $\zeta(2,2) = 3\zeta(4)/4$. The iterated-integral substitution $t \mapsto 1-t$ gives $\zeta(2,1,1) = \zeta(4)$. Expanding the three harmonic numerators in strict sums now gives exactly the displayed values. The first, weight-three formula follows from the same shuffle/stuffle method or from the elementary symmetric integral $\int_0^1 \log x \log(1-x)/(x(1-x)) dx = 2\zeta(3)$.

The decisive finite difference is

$$r_n - r_{n-1} = \frac{12a_{n-1}}{n} + \frac{4}{n^3}. \quad (5.10)$$

It follows by substituting $b_n = b_{n-1} + n^{-1}$, $c_n = c_{n-1} + n^{-2}$, and $e_n = e_{n-1} + n^{-3}$; no limiting argument is involved. Thus finite summation by parts gives

$$\sum_{n=1}^N r_n \log \left(1 + \frac{1}{n} \right) = r_N \log(N+1) - 12 \sum_{n=1}^N \frac{a_{n-1} \log n}{n} - 4 \sum_{n=1}^N \frac{\log n}{n^3}. \quad (5.11)$$

We also need the constant term in $\sum_{n=1}^N r_n/n$. Here is an exact finite identity that makes this normalization checkable. Put $X = H_N$, $Y = H_N^{(2)}$, $Z = H_N^{(3)}$, $W = H_N^{(4)}$, and

$$S_{12} = \sum_{n=1}^N \frac{H_n}{n^2}, \quad S_{13} = \sum_{n=1}^N \frac{H_n}{n^3}, \quad S_{22} = \sum_{n=1}^N \frac{H_n^{(2)}}{n^2}, \quad S_{112} = \sum_{n=1}^N \frac{H_n^2}{n^2}.$$

Then

$$\begin{aligned} \sum_{n=1}^N \frac{r_n}{n} = & -X^4 + 4\gamma X^3 + (-6\gamma^2 + 6\zeta(2))X^2 + (4\gamma^3 - 12\gamma\zeta(2) + 4\zeta(3))X \\ & + 6X^2Y - 12\gamma XY - 4XZ + (-6\gamma^2 + 6\zeta(2))Y - 16\gamma Z - 11W \\ & + 24\gamma S_{12} - 12S_{112} + 20S_{13} + 6S_{22}. \end{aligned} \quad (5.12)$$

For example this identity follows by taking finite differences of H_n^4 , H_n^3 , $H_n^2 H_n^{(2)}$, $H_n H_n^{(2)}$, and $H_n H_n^{(3)}$, then collecting terms. Both sides vanish at $N = 0$, and their exact finite differences are r_N/N .

Let $L = \log N$. Using (5.9) in (5.12), together with $H_N = L + \gamma + O(N^{-1})$, gives

$$\sum_{n=1}^N \frac{r_n}{n} = -L^4 + 12\zeta(2)L^2 + C + o(1), \quad C = \gamma^4 - 18\gamma^2\zeta(2) + 32\gamma\zeta(3) - \frac{23}{2}\zeta(4). \quad (5.13)$$

Also $r_N \log(N+1) = -4L^4 + 24\zeta(2)L^2 + o(1)$. Since $a_{n-1} = 2\zeta(2) - \log^2 n + O(\log n/n)$, the limit

$$A = \lim_{N \rightarrow \infty} \left\{ \sum_{n=1}^N \frac{a_{n-1} \log n}{n} - \zeta(2)L^2 + \frac{L^4}{4} \right\} \quad (5.14)$$

exists. Equations (5.7), (5.11), and (5.13) now imply

$$Q_4 = 18\zeta(4) + 12\zeta(3) - 12\gamma\zeta(2) + 4\zeta'(3) - 12A. \quad (5.15)$$

Every divergent polynomial has been retained until this cancellation.

5.3 Convergent arithmetic and the Laurent coefficient

At a positive integer,

$$a_{n-1} = 2\zeta(2) - \psi(n)^2 - \psi'(n).$$

Moreover $\psi(n)^2 + \psi'(n) - \log^2 n = O(\log n/n)$. Using the standard Stieltjes defining limits in (5.14) gives

$$A = 2\zeta(2)\gamma_1 - \gamma_3 - \sum_{n=1}^{\infty} \frac{(\psi(n)^2 + \psi'(n) - \log^2 n) \log n}{n}.$$

The final series is absolutely convergent because its summand is $O(\log^2 n/n^2)$. Substitution proves (5.4).

To identify the harmonic coordinate, introduce

$$F(s) = \sum_{n=1}^{\infty} \frac{\psi(n)^2 + \psi'(n) - \log^2 n}{n^s}, \quad \Re s > 0.$$

This series and every fixed spectral derivative converge normally on compact subsets of the stated half-plane. Expanding the harmonic numerator in strict sums gives, initially for $\Re s > 1$,

$$F(s) = 2Z_3(s, 1, 1) - 2\gamma Z_2(s, 1) + (\gamma^2 + \zeta(2))\zeta(s) - \zeta''(s). \quad (5.16)$$

This also supplies the required local continuation. The coefficient of u at $s = 1 + u$ is

$$F'(1) = 2\beta_1 - 2\gamma\eta - (\gamma^2 + \zeta(2))\gamma_1 + \gamma_3.$$

On the other hand it is the negative of the convergent sum in (5.4). Inserting this relation proves (5.3). Combining it with the cubic bridge proves (5.5).

5.4 A Mellin normalization for every height-one coefficient

The following classical Mellin mechanism (compare the harmonic-zeta framework in [12]) supplies an independent realization of the coefficient used above. Its all-depth formulation also states precisely where new global information first enters. Let $d \geq 1$, and write

$$Z_d(s, \{1\}^{d-1}) = \sum_{n=1}^{\infty} \frac{H_{n-1}(\{1\}^{d-1})}{n^s}, \quad \frac{1}{\Gamma(1+u)} = \sum_{j \geq 0} c_j u^j.$$

An empty harmonic word has value one. For $t > 0$ put

$$B(t) = -\log(1 - e^{-t}), \quad V_d(t) = B(t)^d - \mathbf{1}_{0 < t < 1} (-\log t)^d.$$

Since $V_d(t) = O(t(1 + |\log t|^{d-1}))$ at zero and decays exponentially at infinity, every logarithmic moment of $V_d(t)/t$ converges absolutely.

Lemma 5.3 (All height-one finite parts and regular coefficients). *For $|u| < 1$, as an identity of meromorphic functions,*

$$\Gamma(1+u)Z_d(1+u, \{1\}^{d-1}) = \frac{1}{u^d} + \frac{u}{d!} \int_0^\infty t^{u-1} V_d(t) dt. \quad (5.17)$$

Consequently the principal part and constant coefficient together are $\sum_{j=0}^d c_j u^{j-d}$. For every $k \geq 1$, the exact regular coefficient is

$$[u^k]Z_d(1+u, \{1\}^{d-1}) = c_{d+k} + \frac{1}{d!} \sum_{j=1}^k \frac{c_{k-j}}{(j-1)!} \int_0^\infty \frac{V_d(t) \log^{j-1} t}{t} dt. \quad (5.18)$$

In particular the finite part is c_d , while the coefficient of u is $c_{d+1} + d!^{-1} \int_0^\infty V_d(t) dt/t$.

Proof. The elementary multiple-polylogarithm identity $\text{Li}_{\{1\}^{d-1}}(z) = (-\log(1-z))^{d-1}/(d-1)!$ gives, first for $\Re s > 1$,

$$\Gamma(s)Z_d(s, \{1\}^{d-1}) = \int_0^\infty t^{s-1} \frac{B(t)^{d-1}}{(d-1)!(e^t - 1)} dt.$$

Subtract $(-\log t)^{d-1}/((d-1)!t)$ on $(0, 1)$. Its Mellin integral at $s = 1 + u$ is u^{-d} . The subtracted kernel is $-V_d'(t)/d!$; no jump term occurs at $t = 1$, since $(-\log 1)^d = 0$. Integrating that remainder by parts produces exactly the integral in (5.17). The boundary terms vanish for $\Re u > -1$, because of the bounds on V_d . Normal convergence of that integral and all its spectral derivatives proves continuation and coefficient extraction. $\square$

For the present depth-three coordinate this yields the particularly simple convergent formula

$$\beta_1 = \frac{\gamma^4}{24} - \frac{\gamma^2 \zeta(2)}{4} + \frac{\gamma \zeta(3)}{3} + \frac{\zeta(4)}{16} + \frac{1}{6} \int_0^\infty \frac{V_3(t)}{t} dt. \quad (5.19)$$

Likewise $\eta = c_3 + \frac{1}{2} \int_0^\infty V_2(t) dt/t$. These formulas evaluate the harmonic coefficients through elementary convergent kernels; they do not identify those kernels with finite combinations of ordinary constants. They also confirm the constant coefficient displayed in (5.2) directly.

5.5 A normalized primitive and every differentiated identity

Put

$$d_0 = 6\gamma^2 - 4\zeta(2), \quad r_0 = 4\gamma^3 - 12\gamma\zeta(2) + 4\zeta(3).$$

For $\Re z > 0$ with the principal logarithm, define

$$\begin{aligned} \mathcal{P}_4(z) = & -\frac{1}{3z^3} - \frac{2\gamma}{z^2} - \frac{d_0}{z} + r_0 \log z + C_* z \\ & + \sum_{n=1}^{\infty} \left[\frac{1}{3n^3} - \frac{1}{3(n+z)^3} - \frac{z}{n^4} \right. \\ & \quad \left. - 4b_n \left\{ \frac{1}{2n^2} - \frac{1}{2(n+z)^2} - \frac{z}{n^3} \right\} \right. \\ & \quad \left. + d_n \left\{ \frac{1}{n} - \frac{1}{n+z} - \frac{z}{n^2} \right\} + r_n \left\{ \log \left(1 + \frac{z}{n} \right) - \frac{z}{n} \right\} \right]. \end{aligned} \quad (5.20)$$

The normally convergent braces are the integrals, vanishing at zero, of the subtracted principal parts. Consequently

$$\mathcal{P}'_4(z) = \psi(z)^4, \quad \text{FP} \int_0^b \psi(x)^4 dx = \mathcal{P}_4(b) \quad (b > 0), \quad \mathcal{P}_4(1) = Q_4.$$

The constant term of $\mathcal{P}_4$ at the singular origin is zero; this fixes the integration constant without an unspecified branch term.

For every integer $m \geq 1$, differentiation of (5.6) gives the further exact identity

$$\begin{aligned} & \sum_{m_1+m_2+m_3+m_4=m} \frac{m!}{m_1!m_2!m_3!m_4!} \prod_{j=1}^4 \psi^{(m_j)}(z) \\ & = (-1)^m m! \sum_{n=0}^{\infty} \left\{ \frac{\binom{m+3}{3}}{(n+z)^{m+4}} - \frac{4b_n \binom{m+2}{2}}{(n+z)^{m+3}} + \frac{(m+1)d_n}{(n+z)^{m+2}} + \frac{r_n}{(n+z)^{m+1}} \right\}. \end{aligned} \quad (5.21)$$

The convergence is normal off the digamma poles, since the slowest term is $O(\log^3 n/n^{m+1})$. At $z = 1$, the left side is a finite polynomial in γ and ordinary zeta values.

Independent checks. The exact checking program verifies the principal parts, the residue finite difference, the full finite identity (5.12), and the cancellations producing (5.3). A separate numerical calculation integrates a locally Taylor-subtracted digamma fourth power and compares it with the convergent arithmetic formula, whose tail is evaluated by Bernoulli asymptotics and Hurwitz-zeta derivatives. At working precision 80 digits, the arithmetic-tail evaluation differs from quadrature by less than $3.4 \cdot 10^{-62}$. Independent Mellin evaluations of η and β_1 , inserted into the main identity, differ from the same quadrature by less than $2.1 \cdot 10^{-74}$. The common value begins

$$Q_4 = 13.5101203760158274116062407128304484419327985333019186009860964 \dots$$

These are numerical diagnostics, not certified enclosures or substitutes for the proof.

Next precise targets. The theorem determines β_1 from the cubic and quartic finite parts, but does not express it in a prescribed algebra of ordinary constants. For the fifth power, the residue polynomial has degree four in the finite harmonic data, and its first difference should produce both height-one depth-four Laurent data and additional harmonic numerators. The correct next task is to derive that exact difference and identify the minimal *stated* family of Laurent coefficients. A second direction is a common Hurwitz shift: the shifted strict harmonic numerator and the integration endpoint must be normalized together, since replacing n by $n + a$ only in the outer denominator does not shift the harmonic prefix or the finite-part coordinate.

6 An ordinary convergent generator for symmetric Tornheim jets

The incoming *Harmonic Parity and Resolvent Identities* report already determines the quartic directional layer and the formal all-order kernel. Its Question R10 asks for additional analytic information about the symmetric quadratic germ controlling cyclic fifth derivatives. We answer the stated integral version of that question. We first give an explicit, normally convergent integral for its full symmetric generating germ. We then extract one transverse quadratic coefficient, including its full finite normalization. Together with the diagonal fifth derivative this coefficient determines every cyclic fifth derivative.

No claim that either retained coefficient has a finite evaluation in ordinary zeta constants is made. The earlier quartic reconstruction, formal dimension count, and cyclic factorization are used with their existing attribution [2]. The holomorphy of the cyclic Tornheim sum near the origin is also a special case of Nakamura's symmetric desingularization theorem [13, Theorem 1.2].

6.1 The fixed holomorphic remainder

Write

$$T(A, B, C) = \sum_{m, n \geq 1} \frac{1}{m^A n^B (m + n)^C}, \quad \mathfrak{g}(C) = \frac{1}{\Gamma(1 + C)}, \quad \Sigma = A + B + C.$$

The series is first used where it converges absolutely; subsequently T denotes its meromorphic continuation. We use exactly the normalization of the preceding report:

$$T(A, B, C) = \zeta(A)\zeta(B) - \frac{C\zeta(B-1)}{A+C} - \frac{C\zeta(A-1)}{B+C} + CR(A, B, C), \quad (6.1)$$

where R is holomorphic and symmetric in A, B near the origin. Here and below an apparent quotient of holomorphic germs by a linear form is assigned its holomorphic extension when its numerator is divisible by that form.

The finite elementary kernel needed below is

$$\begin{aligned} E(A, B, C) = \mathfrak{g}(C) & \left\{ \frac{\Gamma(1-A)\Gamma(1-B)}{\Sigma-2} + \frac{\Gamma(1-A)\zeta(B)}{A+C-1} + \frac{\Gamma(1-B)\zeta(A)}{B+C-1} \right\} \\ & + \zeta(A)\zeta(B) \frac{\mathfrak{g}(C)-1}{C} \\ & - \zeta(B-1) \frac{\Gamma(1-A)\mathfrak{g}(C)-1}{A+C} - \zeta(A-1) \frac{\Gamma(1-B)\mathfrak{g}(C)-1}{B+C}. \end{aligned} \quad (6.2)$$

It is jointly holomorphic near zero. For example, $\Gamma(1-A)\mathfrak{g}(-A) = 1$, proving removability at $A+C=0$.

For completeness, put $F_A(x) = \text{Li}_A(e^{-x})$ and

$$\begin{aligned} S(A, B; x) = & \Gamma(1-A)\Gamma(1-B)x^{A+B-2} + \Gamma(1-A)\zeta(B)x^{A-1} + \Gamma(1-B)\zeta(A)x^{B-1} \\ & - \Gamma(1-A)\zeta(B-1)x^A - \Gamma(1-B)\zeta(A-1)x^B + \zeta(A)\zeta(B). \end{aligned} \quad (6.3)$$

Jonquière's expansion [6, 25.12.12] gives the normally convergent integral

$$\begin{aligned} I_0(A, B, C) = & \int_0^1 x^{C-1} (F_A(x)F_B(x) - S(A, B; x)) dx \\ & + \int_1^\infty x^{C-1} F_A(x)F_B(x) dx \end{aligned} \quad (6.4)$$

and the identity

$$R(A, B, C) = \mathfrak{g}(C)I_0(A, B, C) + E(A, B, C). \quad (6.5)$$

Indeed, integrate the six subtracted terms explicitly in the Mellin representation for T , multiply by $\mathfrak{g}(C)$, and collect the two polar terms of (6.1). This gives (6.2) term by term. In a sufficiently small closed parameter polydisc the omitted Jonquière terms, after multiplication by x^{C-1} , are bounded by $x^{-\delta}(1+|\log x|^N)$ for some $\delta < 1$, including any prescribed finite number of parameter derivatives. At infinity there is uniform exponential decay. These estimates justify the joint holomorphy and all differentiations used below.

6.2 An explicit integral for the symmetric germ

The earlier factorization defines the unique symmetric holomorphic germ J by

$$\begin{aligned} \sum_{\text{cyc}} T(A, B, C) = & 2(\zeta(A)\zeta(B) + \zeta(B)\zeta(C) + \zeta(C)\zeta(A)) \\ & - \zeta(A+B) - \zeta(B+C) - \zeta(C+A) \\ & + 2(\zeta(A) + \zeta(B) + \zeta(C)) + 1 + ABCJ(A, B, C). \end{aligned} \quad (6.6)$$

We retain this normalization exactly. The next formula expresses J through ordinary polylogarithms, Gamma and zeta functions, and ordinarily convergent integrals, without any Tornheim derivative on the right-hand side.

For a holomorphic function H , define

$$\mathcal{D}_{A,B}H(A, B, C) = \frac{H(A, B, C) - H(A, 0, C) - H(0, B, C) + H(0, 0, C)}{AB}.$$

At zero parameters this means the corresponding derivative. Set

$$\begin{aligned} d_A(x) &= \frac{\text{Li}_A(e^{-x}) - \text{Li}_0(e^{-x})}{A}, & g_A(x) &= \frac{\Gamma(1-A)x^A - 1}{A}, \\ p_A &= \frac{\zeta(A) + 1/2}{A}, & q_A &= \frac{\zeta(A-1) + 1/12}{A}, \end{aligned}$$

with the holomorphic values at $A = 0$. Define

$$\begin{aligned} S_\Delta(A, B; x) = & \frac{g_A(x)g_B(x)}{x^2} + \frac{g_A(x)p_B + g_B(x)p_A}{x} \\ & - g_A(x)q_B - g_B(x)q_A + p_Ap_B, \end{aligned} \quad (6.7)$$

$$\begin{aligned} I_\Delta(A, B, C) = & \int_0^1 x^{C-1} (d_A(x)d_B(x) - S_\Delta(A, B; x)) dx \\ & + \int_1^\infty x^{C-1} d_A(x)d_B(x) dx. \end{aligned} \quad (6.8)$$

Theorem 6.1 (Ordinary convergent symmetric generator). *The integrals (6.8) converge normally in a common neighborhood of $(0, 0, 0)$, including after arbitrary fixed parameter derivatives, and*

$$J(A, B, C) = \sum_{\text{cyc}} \{g(C)I_{\Delta}(A, B, C) + \mathcal{D}_{A,B}E(A, B, C)\}. \quad (6.9)$$

Proof. Apply the rectangular difference in A, B to (6.4). The difference of the product $F_A F_B$, divided by AB , is $d_A d_B$; the corresponding difference of S is exactly S_{Δ} . Cauchy estimates on a slightly larger parameter polydisc preserve the integrable bounds of (6.4); equivalently, one can use $d_A(x) = g_A(x)/x + p_A - q_A x + O(x^{2-\epsilon})$, locally uniformly with logarithmic factors after differentiation. Thus $I_{\Delta} = \mathcal{D}_{A,B}I_0$, with normal convergence.

In the cyclic sum of (6.1), the two terms with denominator $A + C$ combine to $-\zeta(B - 1)$, and similarly for the other denominators. All remaining explicit zeta terms depend on at most two variables. The complete rectangular three-variable difference therefore annihilates them and gives

$$ABCJ(A, B, C) = \sum_{\text{cyc}} C(R(A, B, C) - R(A, 0, C) - R(0, B, C) + R(0, 0, C)).$$

To check the left-hand side, apply the same difference to (6.6): all its explicit terms vanish, and $ABCJ$ vanishes on every coordinate plane. Divide by ABC , substitute (6.5), and use holomorphic extension at zero parameters. This proves (6.9). $\square$

The theorem is an analytic representation of the previously defined germ. It supplies all symmetric jet coefficients in one fixed, ordinary convergent kernel. Its content includes the endpoint subtractions and their finite normalization; replacing the kernel by an unsubtracted integral would not preserve its Taylor coefficients.

6.3 The first transverse quadratic coefficient

Write γ for Euler's constant and put

$$L = \log(2\pi), \quad z_j = \zeta^{(j)}(0), \quad y_j = \zeta^{(j)}(-1), \quad \kappa_2 = \zeta(2), \quad \kappa_4 = \zeta(4).$$

The quadratic homogeneous part of J has the form

$$J_2(A, B, C) = j_{20}(A^2 + B^2 + C^2) + \chi(AB + BC + CA). \quad (6.10)$$

Thus $\chi = [AB]J$ is the additional scalar to be determined. For $x > 0$, let

$$f_j(x) = \partial_s^j \text{Li}_s(e^{-x}) \Big|_{s=0}, \quad h(x) = \log x + \gamma,$$

and define

$$P_{21}(x) = \frac{h(h^2 + \kappa_2)}{x^2} + \frac{z_1(h^2 + \kappa_2) + z_2 h}{x} - y_1(h^2 + \kappa_2) - y_2 h + z_1 z_2, \quad (6.11)$$

$$P_{22}(x) = \frac{(h^2 + \kappa_2)^2}{x^2} + \frac{2z_2(h^2 + \kappa_2)}{x} - 2y_2(h^2 + \kappa_2) + z_2^2, \quad (6.12)$$

$$\mathcal{K}(x) = \frac{1}{4}f_2(x)^2 + h(x)f_2(x)f_1(x), \quad P(x) = \frac{1}{4}P_{22}(x) + h(x)P_{21}(x). \quad (6.13)$$

Let

$$\mathcal{I} = \int_0^1 \frac{\mathcal{K}(x) - P(x)}{x} dx + \int_1^\infty \frac{\mathcal{K}(x)}{x} dx. \quad (6.14)$$

Both integrals converge ordinarily and absolutely.

Theorem 6.2 (Explicit transverse fifth-order coordinate). *The coefficient in (6.10) is*

$$\chi = \mathcal{I} + \mathcal{E}, \quad (6.15)$$

where

$$\begin{aligned} \mathcal{E} = & -\frac{10\gamma^4 + 20\gamma^3 + 12\gamma^2\kappa_2 + 30\gamma^2 + 12\gamma\kappa_2 + 30\gamma + 2\kappa_2^2 + 6\kappa_2 + 15}{16} \\ & + \frac{-\gamma^4 - 2\gamma^2\kappa_2 + \kappa_2^2 + 2\kappa_4}{4} y_1 - \frac{\gamma(\gamma^2 + \kappa_2)}{2} y_2 \\ & - (\gamma^3 + 3\gamma^2 + 6\gamma + 6 + (\gamma + 1)\kappa_2) z_1 - \frac{3\gamma^2 + 6\gamma + 6 + \kappa_2}{2} z_2 \\ & + \frac{\gamma^2 - \kappa_2}{2} z_1 z_2 + \frac{\gamma}{4} z_2^2. \end{aligned} \quad (6.16)$$

Proof. All derivatives of R in this proof are evaluated at the origin. The coefficient of $A^2 B^2 C$ in $\sum_{\text{cyc}} CR(A, B, C)$ is

$$\frac{1}{4}R_{AABB} + R_{AABC}.$$

The first cyclic term contributes $R_{AABB}/4$, and the other two contribute $R_{AABC}/2$ each. The explicit zeta terms in (6.6) and the cyclic form of (6.1) have no monomial involving all three variables. Consequently

$$\chi = \frac{1}{4}R_{AABB} + R_{AABC}. \quad (6.17)$$

Differentiate (6.5), using $\mathfrak{g}(0) = 1$ and $\mathfrak{g}'(0) = \gamma$. The integral contribution to (6.17) is precisely (6.14): $S_{AAB} = P_{21}$, $S_{AABB} = P_{22}$, and the derivative of $\mathfrak{g}(C)x^{C-1}$ at zero is $h(x)/x$. For a direct convergence check, the differentiated Jonquière expansions are

$$f_1(x) = \frac{h}{x} + \sum_{k \geq 0} \frac{(-1)^k \zeta'(-k)}{k!} x^k, \quad f_2(x) = \frac{h^2 + \kappa_2}{x} + \sum_{k \geq 0} \frac{(-1)^k \zeta''(-k)}{k!} x^k.$$

The displayed subtractions remove both singular products, the cross terms of analytic degrees zero and one, and the constant analytic product. Hence $(\mathcal{K} - P)/x = O(1 + |\log x|^3)$ at zero. At infinity $f_1, f_2 = O(e^{-2x})$, so $\mathcal{K}/x = O((1 + \log x)e^{-4x}/x)$.

The remaining finite contribution is

$$\mathcal{E} = [A^2 B^2]E + 2[A^2 BC]E.$$

Insert

$$\begin{aligned} \log \Gamma(1 - A) &= \gamma A + \frac{\kappa_2}{2} A^2 + \frac{\zeta(3)}{3} A^3 + \frac{\kappa_4}{4} A^4 + O(A^5), \\ \log \mathfrak{g}(C) &= \gamma C - \frac{\kappa_2}{2} C^2 + \frac{\zeta(3)}{3} C^3 - \frac{\kappa_4}{4} C^4 + O(C^5) \end{aligned}$$

into the elementary expression (6.2). Expand its three nonresonant denominators geometrically and divide the three removable quotients as holomorphic power series. Extraction of the two coefficients gives (6.16). This is finite polynomial algebra; the accompanying exact script performs it independently with all zeta derivatives left symbolic. $\square$

6.4 Every cyclic fifth derivative

Put $\Omega_5 = W_{1,1,1}^{(5)}(0)$, where $W_{a,b,c}(t) = T(at, bt, ct)$ is restricted before taking its one-variable continuation at zero. Write

$$e_1 = a + b + c, \quad e_2 = ab + bc + ca, \quad e_3 = abc, \quad D = e_1^2 - 3e_2, \quad p_2 = e_1^2 - 2e_2.$$

Theorem 6.3 (Cyclic fifth-order formula). *If the three pairwise sums of a, b, c are nonzero, then*

$$\begin{aligned} \sum_{\text{cyc}} W_{a,b,c}^{(5)}(0) &= e_3 p_2 \Omega_5 - 120 e_3 D \chi - 5(e_1 e_2 - 3e_3) D L z_4 \\ &\quad + 20(e_1 e_2^2 - 4e_1^2 e_3 + 3e_2 e_3) z_2 z_3 \\ &\quad + (-2e_1^5 + 5e_1^3 e_2 - 5e_1 e_2^2 + 47e_1^2 e_3 - 69e_2 e_3) z_5. \end{aligned} \quad (6.18)$$

For arbitrary complex slopes the same polynomial gives the fifth derivative of the holomorphic cyclic sum restricted as a whole. This last assertion does not assign values to its individual singular summands.

Proof. On the diagonal, differentiation of (6.6) gives

$$j_{20} + \chi = \frac{\Omega_5 + 10Lz_4 - 40z_2 z_3 + 32z_5}{120}.$$

Thus (6.10) is completely determined by Ω_5 and the ordinarily convergent coordinate (6.15). For general slopes, the fifth derivative of the explicit part of (6.6) is

$$-5Lz_4 \sum_{\text{sym}} a^4 b + 20z_2 z_3 \sum_{\text{sym}} a^3 b^2 - ((a+b)^5 + (b+c)^5 + (c+a)^5) z_5.$$

The remaining derivative is $120abcJ_2(a, b, c)$. Substitute the preceding value of j_{20} , use the elementary symmetric polynomial identities, and collect terms. This yields (6.18). Since (6.6) is jointly holomorphic, its cyclic restriction exists even when an individual pairwise divisor would prevent restricting a single Tornheim term. $\square$

For instance, the explicit asymmetric identity is

$$\begin{aligned} W_{1,2,3}^{(5)}(0) + W_{2,3,1}^{(5)}(0) + W_{3,1,2}^{(5)}(0) - 84\Omega_5 \\ = -2160\chi - 720L\zeta^{(4)}(0) + 1200\zeta''(0)\zeta'''(0) - 1704\zeta^{(5)}(0). \end{aligned} \quad (6.19)$$

For real slopes, $D = \frac{1}{2}((a-b)^2 + (b-c)^2 + (c-a)^2)$. Consequently a single real cyclic fifth ray with $abc \neq 0$ eliminates the displayed transverse coefficient only on the diagonal. For complex slopes the cone $D = 0$ has additional points. These are coefficient statements in the specified two coordinate representation, not claims of arithmetic independence.

6.5 Numerical checks and the remaining arithmetic problem

The package evaluates $\mathcal{I}$ using differentiated Jonquière series on $(0, 1)$ and exponentially convergent incomplete-Gamma sums on $(1, \infty)$. At 70 decimal digits of working precision the diagnostic values are

$$\begin{aligned}\mathcal{I} &= -0.0128358838132664551878307355997471413775071472849\dots, \\ \mathcal{E} &= 18.6522612153652911636439613245591942979227651526949\dots, \\ \chi &= 18.6394253315520247084561305889594471565452580054100\dots, \\ \Omega_5 &= 8808.87952093843854353140320772504731742914386526227\dots\end{aligned}$$

An independently evaluated Mellin continuation of individual T rays checks (6.18) on the cycles generated by $(1, 2, 3)$, $(1, 1, 2)$, and $(1, -2, 3)$; the largest recorded discrepancy is below $6 \cdot 10^{-30}$. These are uncertified numerical diagnostics, not interval enclosures or proofs of the analytic identities. The exact coefficient script checks the elementary correction and the full symbolic fifth-order formula.

The new generator answers the ordinary-integral alternative of Question R10 and makes all higher symmetric coefficients concrete. Finite ordinary-constant evaluations of χ and Ω_5 remain open. A sharper next problem is to find an independent Gamma or Barnes representation of the kernel in (6.14), with its normalization (6.16) unchanged. At the following order, the same generator yields the three coefficients of the symmetric cubic germ J_3 ; any reduction below that formal count requires an additional identity.

7 Source audit, verification, and further research

7.1 A precise correction to a published Barnes identity

The inspection of the Barnes Fourier source revealed a pointwise algebraic error in equation (7) of the inspected arXiv version of Mezó's paper [11]. That equation identifies $\log^2 G(x)$ with

$$\frac{1}{4} \left\{ \log^2(G(x)G(1-x)) + \log^2(G(x)/G(1-x)) \right\}.$$

On $(0, 1)$ both Barnes values are positive. Put $a = \log G(x)$ and $b = \log G(1-x)$. The displayed expression is $(a^2 + b^2)/2$, not a^2 . Thus the corrected pointwise identity is

$$\boxed{\frac{\log^2 G(x) + \log^2 G(1-x)}{2} = \frac{1}{4} \left\{ \log^2(G(x)G(1-x)) + \log^2(G(x)/G(1-x)) \right\}}. \quad (7.1)$$

Alternatively, to keep the original left side one must add

$$\frac{1}{2} \log(G(x)G(1-x)) \log(G(x)/G(1-x))$$

to the original right side.

This is an exact objection, not a small numerical discrepancy. From $G(1+x) = \Gamma(x)G(x)$ and $G(1) = 1$ one gets $G(x) = x(1 + O(x))$ as $x \downarrow 0$, while $\log G(1-x) = O(x)$. Therefore the original left side has leading term $\log^2 x$, whereas its stated right side has leading term $\frac{1}{2} \log^2 x$.

The distinction has a narrow consequence. After integration over $(0, 1)$, reflection makes $\int \log^2 G(1-x) = \int \log^2 G(x)$, and the added product is odd under $x \mapsto 1-x$. The integrated equality used next in that paper is valid. We therefore propose correcting the pointwise line or placing its equality after integration; we do not infer that its subsequent integrated theorem is false. The source version and location are recorded in `integration/CORRECTIONS.md`.

7.2 What the repository audit does and does not establish

The source inventory contains 81 manuscript chapter files and all 11 incoming archives present at the pinned commit. The detailed mathematical review concentrates on the conjecture status records, the cotangent-resolvent programme, the harmonic power bridges, and the cyclic Tornheim remainder. The inventory is not a claim that every theorem in all these documents has received a fresh proof audit.

The following distinctions are necessary for correct integration.

1. The classical Barnes/Hurwitz conversion, the log- G Fourier coefficients, balanced negapolygamma primitives, beta-integral binomial identities, and symmetric Tornheim desingularization keep their existing attribution. The new contribution is the proved resolvent and coefficient reduction with the stated normalization.
2. The earlier report already contains the quartic directional Tornheim reconstruction and the formal cyclic dimension count. Section 6 supplies the requested explicit convergent integral, and then the fifth-order formula using that analytic coordinate; it does not relabel the earlier formal result as new.
3. The quartic digamma moment in Section 5 is a different question from the quartic directional Tornheim layer. Its depth-three coefficient β_1 is strict, as specified in (5.2). An inclusive harmonic coefficient cannot be substituted without its diagonal correction.
4. At a resonance, a Laurent constant in the spectral or kernel exponent is not automatically the finite part in the original integration coordinate. Equations (2.22) and (4.15) explicitly perform the conversion.
5. The current S_6 and revised S_8 coefficient vectors remain conjectural. The pre-existing rejection of the old frozen S_8 vector must retain its separate label and status. No present decimal evaluation changes any of these statuses.

Within the inspected repository material we found no additional mathematical error that is established by the present analysis. The pointwise Barnes correction above is to the cited external source. The attached claim ledger distinguishes proved identities, classical ingredients, numerical diagnostics, and open questions. The package consists of research documents and reproducible calculations, not machine-checked ProveIt proof objects.

7.3 Independent checks and reproducibility

Four independent lines of calculation accompany the proofs. The Barnes checks evaluate the depth-two derivative from its own double series and evaluate the Fourier sum $\sum q^n T_n$ separately. The same code compares the complete transform with direct log G quadrature in both half-planes, and checks the removable $q = 0$ real component. The polynomial Hurwitz checks compare the subtracted double series and its spectral derivative with direct Hurwitz quadrature, including a weighted Stieltjes coefficient.

The complex-power checks compare its Mellin generator with the original integral at nonintegral complex parameters, and independently test its Gamma closure and a noncommuting finite-part example. The quartic checks use three routes: local digamma subtraction, the convergent harmonic series with a zeta-derivative tail, and the independent Mellin coefficients η, β_1 . The cyclic fifth

checks compare the new ordinary-integral coordinate with an independently evaluated continuation of three asymmetric cycles. The pre-existing evaluator used in that comparison is preserved with its provenance.

Finite exact algebra checks cover the multiple-Gamma coefficient polynomials and normalization, quartic principal parts and their finite summation identity, the elementary Tornheim correction, and the symbolic fifth-order polynomial. These calculations are useful for detecting signs, factorials, and omitted constants. The analytic assertions—normal convergence, uniqueness of an entire remainder, branches, continuation, and the permitted coefficient interchanges—are proved in the text.

All numerical records are explicitly non-interval diagnostics. Their working precision and truncations are recorded. Several independent values agreeing to many digits cannot certify an integer relation or a finite closed form. Scripts, dependency versions, replay commands, numerical records, and a file-hash manifest are included so that integration can preserve both the theorem and its actual verification status.

7.4 Further research questions

The next problems are formulated as exact identities or reductions in specified function classes. The first four build most directly on the new formulas.

F1. Products of individual Barnes logarithms. Determine the minimal explicit nested-series class for

$$\int_0^1 W(x) \log \Gamma_r(x) \log \Gamma_t(x) R_z(x) dx.$$

The polynomial Hurwitz decomposition reduces this to a bilinear weighted transform. The real question is whether its lattice splitting closes at depth three, with all diagonal corrections explicit, and which reflection combinations reduce the depth.

F2. A second representation of $\mathcal{D}(q)$. Find a Gamma, Barnes, or iterated-integral expression for (3.13) that is independent of its defining depth-two sum. A useful target is an exact differential or functional equation under $q \mapsto q/(q-1)$, with a specified branch domain and initial value. Establish whether it reduces to ordinary polylogarithms and their order derivatives; do not infer such a reduction from a successful numerical fit at isolated points.

F3. All higher digamma powers. Starting from the principal parts of $\psi(z)^5$, derive the exact finite difference of its residue polynomial. Determine which height-one depth-four Laurent coefficients and which additional harmonic numerators survive summation by parts. A satisfactory result must state a finite family of retained coefficients and fix the entire remainder and every finite-part constant, as was done for the fourth power here.

F4. Shifted quartic harmonic bridges. For $a > 0$, derive a common-shift analogue of (5.3), using prefixes $\sum_{k=0}^{n-1} (k+a)^{-r}$ and the corresponding outer denominator. Track simultaneously the origin of the integration coordinate, the lower summation endpoint, and generalized Stieltjes constants $\gamma_j(a)$. Merely shifting the outer denominator leaves a different problem.

F5. Nonpositive kernel resonances. Extend the pointwise finite-part comparison of Theorem 4.4 to $\Re \lambda \leq 0$, keeping contributions from both endpoints. There is already a concrete additional

phase at $\lambda = -1$. The continued mean of the kernel is A^λ , so its value there is $-z - \epsilon\pi i$, whereas

$$\text{FP}_x \int_0^1 R_z(x)^{-1} dx = \text{FP}_x \int_0^1 (K(x) - z) dx = -z.$$

The last equality follows by integrating $K = (\log \sin \pi x)'$ on the symmetric cutoff interval. Thus the positive-resonance rule cannot simply be copied to negative powers.

- F6. Rational kernel exponents.** For a fixed nonintegral rational λ , determine whether $\mathcal{L}_\lambda(u, q)$ admits a finite representation in a specified hypergeometric or multiple-polylogarithm class at integral u . Its Gamma specialization at $q = 0$ and its exact exponent jets are constraints on any proposed reduction. The unrestricted complex- λ Dirichlet series is presently retained as an explicit coordinate.
- F7. The arithmetic of β_1 .** Decide whether the normalized integral in (5.19) reduces to a specified algebra generated by ordinary zeta derivatives, Stieltjes constants, and Gamma or Barnes values at rational arguments. The proved identity for $Q_4 + 4\gamma Q_3$ gives an exact equivalence with a digamma finite-part evaluation, not an arithmetic independence statement. A new reduction must add information beyond that equivalence.
- F8. Higher symmetric Tornheim coefficients.** Use (6.9) to extract the three homogeneous cubic coefficients of J_3 , including their elementary corrections. They control the cyclic sixth-derivative layer. Determine which reflection or differential identities reduce the formal three-coordinate count. For the present quadratic layer, seek a second exact representation of χ that could lead to a finite evaluation of χ or Ω_5 .
- F9. Commensurable Barnes periods.** Extend the individual resolvent theorem from unit periods to positive rationally commensurable periods. Multiplicity sequences should split into finitely many polynomial residue-class pieces, suggesting shifted Hurwitz and root-of-unity polylogarithm transforms. Prove that reduction with the exact normalization; arbitrary incommensurable periods are a separate problem and need not remain at finite depth two.
- F10. Rational shifts and cyclotomic colors.** Combine the polynomial weight with a rational Hurwitz shift, or with a finite Fourier character. Determine the exact colored analogue of $\mathcal{O}_{\alpha, \beta}(q)$ and its residue cancellation. An effective theorem should specify cumulative color products and convergence, since individual arguments of modulus greater than one need not obstruct the coupled nested sum.
- F11. Exact certificates for the current S_6 and S_8 .** Keep each current candidate vector fixed. Try to express its difference from the claimed integral in a finite basis of iterated integrals or a rigorously defined cyclotomic quotient, and reduce that expression by proved shuffle, stuffle, distribution, and path relations. The deliverable should be an exact certificate, or an exact obstruction to the specified reduction, rather than additional matching digits. The already-rejected old S_8 vector is not the current target.
- F12. Formal finite-part and convergence infrastructure.** Formalize a reusable theorem that subtracts a parameter-dependent endpoint jet, integrates its monomials, and compares coefficient extraction with the original-coordinate finite part. Pair it with normal-convergence lemmas for a geometric outer index and a subtracted power-law inner index. These two modules

would support most identities here without treating each moving pole as a separate ad hoc calculation.

7.5 Integration guide

The suggested integration order is the polynomial transform, the individual Barnes theorem, the complex-power generator, the quartic harmonic bridge, and finally the transverse Tornheim generator. The first two share their new analytic input; the remaining branches have their own local subtraction proofs. All manuscript labels in the deliverable use a common prefix. The standalone source and the modular source compile the same article. No external PDFs are bundled: the bibliography links to the primary sources, while the provenance records identify the exact repository snapshot and the reused numerical evaluator.

References

- [1] V. Reshetnikov, *ProveIt: polylogarithm manuscript and incoming research collection*, repository snapshot 7bd45777a8a127e5dc69aff8887ca36a79a3059d, 11 October 2026. [Pinned manuscript](#); [pinned incoming collection](#). The attached inventory lists all source files used for the targeted audit.
- [2] ProveIt research collection, *Harmonic Parity and Resolvent Identities*, incoming report dated 11 October 2026, in [1]. Sections 5 and 7; in particular the local remainder and symmetric germ with labels `qtr:polar`, `qtr:kernel`, `qtr:J`, and questions R4, R6, R10. [Pinned source archive](#).
- [3] ProveIt research collection, *Cubic Tornheim Derivatives and Harmonic Stieltjes Constants*, incoming report dated 11 October 2026, in [1]. The cubic digamma–harmonic bridge and Section 7, question 6. [Pinned source archive](#).
- [4] NIST Digital Library of Mathematical Functions, §25.11, *Hurwitz Zeta Function*. <https://dlmf.nist.gov/25.11>. Fourier expansion, argument shifts, Bernoulli values, and derivatives. Accessed 11 October 2026.
- [5] NIST Digital Library of Mathematical Functions, §§5.7, 5.11, 5.15, 5.17, *Gamma Function*. <https://dlmf.nist.gov/5.15>; <https://dlmf.nist.gov/5.17>. Digamma and polygamma expansions, asymptotics, and the Barnes G function. Accessed 11 October 2026.
- [6] NIST Digital Library of Mathematical Functions, §25.12, equation 25.12.12, *Polylogarithms*. <https://dlmf.nist.gov/25.12.E12>. The Jonquière expansion used in Mellin subtractions. Accessed 11 October 2026.
- [7] O. Espinosa and V. H. Moll, *On some integrals involving the Hurwitz zeta function: Part 2*, Ramanujan Journal **6** (2002), 449–468. https://www.math.tulane.edu/~vhm/papers_html/hurwitz2.pdf.
- [8] O. Espinosa and V. H. Moll, *A generalized polygamma function*, Integral Transforms and Special Functions **15** (2004), 101–115. <https://arxiv.org/abs/math/0305079>.
- [9] V. S. Adamchik, *Multiple Gamma Function and Its Application to Computation of Series*, arXiv:math/0308074 (2003). <https://arxiv.org/abs/math/0308074>. Classical multiple-Gamma normalization and Hurwitz-zeta conversion.

- [10] I. Vardi, *Determinants of Laplacians and multiple gamma functions*, SIAM Journal on Mathematical Analysis **19** (1988), 493–507. <https://doi.org/10.1137/0519035>.
- [11] I. Mező, *The Fourier series of the log-Barnes function*, arXiv:1604.00753v1 (2016). <https://arxiv.org/pdf/1604.00753v1>. Known Fourier coefficients; the pointwise correction in Section 7 concerns equation (7), printed page 6 of this specific version, and preserves its integrated use.
- [12] L. Kargin, A. Dil, M. Cenkci, and M. Can, *On the Stieltjes constants with respect to harmonic zeta functions*, Journal of Mathematical Analysis and Applications **525** (2023), article 127302. <https://arxiv.org/abs/2304.03517>; <https://doi.org/10.1016/j.jmaa.2023.127302>. The harmonic-zeta coefficient framework is prior work; our strict depth-three normalization is specified independently in the text.
- [13] T. Nakamura, *Rapidly convergent series representations of symmetric Tornheim double zeta functions*, arXiv:2103.16873 (2021), Theorem 1.2. <https://arxiv.org/abs/2103.16873>.